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A Systematic Literature Review of Artificial Intelligence Techniques for Security Log Analysis in Modern Distributed and Cloud-Based Systems

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PREPARED BY

[Shaima Abdulkawi Al-Thari]

SUPERVISED BY

[Dr. Ahmed Abdullah Al-Shalabi]

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1 Introduction

1.1 Background

The modern computing environment has undergone significant changes due to the advent of cloud computing, the use of microservice architecture, and large-scale distributed infrastructure. The result has been the constant creation of huge amounts of log information that tracks system activities, user interactions, service interactions, authentication actions, and operations of infrastructure components.

In the modern world, log files are no longer considered an additional debugging tool but rather an essential source of information about system activity, performance issues, and security issues. According to recent research, security logs are now indispensable in security monitoring, forensic investigations, compliance audits, and the detection of threats in large-scale distributed systems [1–3].

Security logs are different from system logs in that they specifically log events related to security, such as authentication, privilege escalation, API activity, audits, and unauthorized access attempts. What makes security logs useful is that they may have early signs of malicious behavior or policy violations. However, the full value can be derived from them only through the correlation of events over time, across different systems, and users [2, 4, 5].

Although they are crucial, current security logs pose certain challenges related to their analysis. The amount of information produced in real-time distributed settings makes it impossible to inspect the data manually. Moreover, the logs themselves come from multiple sources in various formats, schemas, and semantics. As a result, they become difficult to preprocess and analyze together. Furthermore, security-related events usually represent only a small share of all log data, causing extreme class imbalance problems that make traditional analysis methods hard to apply [6–10].

As a solution to this problem, Artificial Intelligence (AI) is currently one of the most important technologies enabling automatic log analysis. Early methods were based on classical machine learning algorithms, which needed handcrafted features for identifying abnormal events in log data. However, with the development of deep learning methods, there appeared models that could learn representations without handcrafted features directly from the raw or semistructured logs. Recently, transformer architectures revealed a high potential to detect long-range dependencies and contexts in the log sequences. Meanwhile, large language models (LLMs) and retrieval-augmented generation (RAG) systems expanded the boundaries of AI-based log analysis through reasoning, context understanding, and support of analysts [R14, R17, R19, R23, R28].

Simultaneously, security log analysis has moved beyond the detection of anomalies. Recent studies tend to pay more attention to higher-level tasks of security, such as threat

hunting, root cause analysis, incident response, and self-healing. It is evident that this tendency is related to a transition from reaction-based security decision-making systems to the proactivity of these systems. Still, however, the area is quite fragmented, and approaches used differ greatly in many aspects.

1.2 Problem Statement

In spite of the fast development of modern infrastructure and its growing tendency towards cloud-native and distributed computing, the analysis of security logs remains one of the unresolved challenges in cybersecurity research. Modern systems are constantly generating huge amounts of diverse logs covering various events, such as system events, authentication actions, API operations, and cross-service communication. Although these logs provide important information regarding system processes and can be used to detect security problems, the diversity, volume, and time complexity of the logs make their analysis impossible using classic methods [1–3].

Over the last ten years, the paradigm of Artificial Intelligence has proven to be an adequate solution to the described challenge. Various approaches have been developed, including classical machine learning, neural network modeling, transformer architecture, and more recent ones, such as large language models (LLMs) and retrieval-augmented generation (RAG) frameworks. However, even taking into account the considerable progress that has been made, the existing literature lacks cohesion from several viewpoints: different learning approaches, goals of the security system, and log data representation [6–10].

The high level of fragmentation in the literature in question makes it problematic to develop a cohesive body of knowledge. In other words, the research results are evaluated in a way that does not make them comparable in many aspects due to the different datasets and criteria used during evaluation. Furthermore, while many studies have been dedicated to anomaly detection in security logs, relatively few studies have explored higher-level security goals, such as root cause analysis, reasoning about threats, responding to incidents, and healing. Such emerging trends are crucial for evolving from reactive security monitoring towards proactive and autonomous security systems. However, the ongoing inability to link detection, interpretation, and response elements together still stands out as a critical issue in the field of research [1, 3, 11–13].

Moreover, modern developments in the fields of transformer-based and large language models allow us to perform contextual reasoning in relation to log data. Yet, such models create numerous challenges associated with their tendency to hallucinate, interpretable nature, domain adaptation, and trustworthy performance in terms of security [9, 10]. Therefore, despite excellent results in general language processing tasks, applicability in real cybersecurity settings remains unknown.

Therefore, the main problem studied in this paper is the lack of systematic, structured,

and holistic synthesis of existing literature on AI-based analysis of security logs that takes methodological perspective, security objectives, and data properties into consideration. This gap significantly limits the ability to systematically compare existing approaches, understand their relative strengths and limitations, identify research trends and evolution patterns, and derive meaningful insights for future research and practical deployment.

1.3 Motivation

Given the rapid development of the IT landscape due to cloud-native architecture, microservices, and other factors related to the emergence of highly distributed systems, security monitoring and observability have been dramatically changed. Under the conditions of the above mentioned contexts, the security logs become one of the most important sources of operational intelligence, as they provide information about detailed traces of system activity, actions of the users, and inter-service communications [1–3].

Despite the importance of the above data, the growing complexity, heterogeneity, and amount of security log data generate significant analytical problems that go beyond the possibilities of the classic approach based on rule-based and manual analysis of the data. In particular, the features of the dynamics and high throughput of modern systems make the real-time detection of meaningful security insights very complicated, especially in cases when malicious activity is rare, sophisticated, and distributed in different systems [7, 9, 14].

As a result, there has been an evident trend to use AI approaches for the automation of security logs analysis. The methods based on machine learning, deep learning architecture, transformers, LLMs, and RAG models have shown a significant potential in solving the above mentioned problems.

However, in spite of these advances, the modern research situation remains highly fragmented, with studies based on different assumptions, datasets, learning frameworks, and evaluation criteria making it impossible to develop an overview of the state of the art in the area. Furthermore, while there have been significant advances in anomaly detection, there is still a gap in the area of incorporating high-level security features, such as root cause analysis, threat reasoning, incident response, and automation of remediation.

Furthermore, this problem becomes even more pronounced given the emergence of transformer and large language model-based approaches to the problem. Although these approaches provide significant semantic understanding and reasoning power, they are associated with issues of reliability, vulnerability to hallucinations, domain adaptation, and trustworthiness in the context of security (see [9, 10]).

As a result, there is a significant need for a systematic and comprehensive review of the current state of knowledge on AI applications in security log analysis. This need stems not only from the need to synthesize the fragmented contributions in the area but

also to compare their merits and limitations.

Therefore, this paper provides the justification for conducting a literature review as a solution to this problem.

1.4 Research Aim

The aim of this study is to systematically review, analyze, and synthesize existing research on Artificial Intelligence-based security log analysis in order to provide a comprehensive and structured understanding of the field. The study seeks to identify, categorize, and evaluate current methodologies across different AI paradigms, including classical machine learning, deep learning, transformer-based models, and large language model (LLM) and retrieval-augmented generation (RAG) frameworks, with respect to their application in security log analysis tasks such as anomaly detection, threat detection, root cause analysis, incident response, and self-healing systems.

Furthermore, this study aims to develop a unified analytical perspective that integrates security objectives, methodological approaches, and log data characteristics, thereby addressing the fragmentation in the existing literature and providing a coherent foundation for future research and practical applications in intelligent security systems.

1.5 Objectives of the Study

The objectives of this study are as follows:

- To systematically identify and classify existing research on AI-based security log analysis across major AI paradigms, including traditional machine learning, deep learning, transformer-based models, and LLM/RAG frameworks.
- To develop a structured taxonomy of security log analysis approaches based on three dimensions: AI techniques, security objectives, and log data sources.
- To critically compare and analyze existing approaches in terms of their methodologies, strengths, limitations, and applicability in different security contexts.
- To investigate the evolution of research trends in security log analysis, highlighting the shift from traditional anomaly detection toward intelligent and autonomous security systems.
- To identify key research gaps and future directions in AI-driven security log analysis, particularly in areas such as root cause analysis, incident response, explainability, and self-healing systems.

1.6 Research Questions

Based on the defined objectives, this study aims to address the following research questions:

- **RQ1:** What are the existing research works on AI-based security log analysis, and how can they be systematically categorized across different AI paradigms such as traditional machine learning, deep learning, transformer-based models, and LLM/RAG frameworks?
- **RQ2:** What are the key dimensions required to construct a comprehensive taxonomy of AI-based security log analysis approaches, considering AI techniques, security objectives, and log data sources?
- **RQ3:** How do existing AI-based approaches compare in terms of their methodologies, strengths, limitations, and applicability to different security log analysis tasks and environments?
- **RQ4:** How has the research landscape of security log analysis evolved over time, particularly in the transition from traditional anomaly detection to intelligent and autonomous security systems?
- **RQ5:** What are the main research gaps and future directions in AI-driven security log analysis, especially in areas such as root cause analysis, incident response, explainability, and self-healing systems?

1.7 Scope and Limitations

This study conducts a Systematic Literature Review (SLR) to provides an analysis of Artificial Intelligence (AI)-driven methods for security log analysis. This SLR is confined to studies applying AI-based approaches to analyzing security logs.

The research covers works based on traditional machine learning, deep learning approaches, transformer architectures, as well as recently emerged LLM and RAG frameworks. These approaches are assessed regarding several security purposes, such as anomaly detection, threat detection, root cause analysis, incident response, explainability, and self-healing systems. Various kinds of security logs, which can be analyzed using the described AI techniques, include application logs, authentication logs, API logs, audit logs, system logs, and logs created in distributed and cloud environments.

To ensure methodological consistency and research quality of achieving methodological homogeneity and high research quality, the SLR focuses on peer-reviewed journal papers and conference papers, which can be found in scientific databases, namely IEEE

Xplore, ACM Digital Library, SpringerLink, and Scopus. Studies written in English are considered.

Several limitations define the boundaries of this study. First, the review excludes research that does not explicitly utilize log data as the primary source of security information. Studies focusing exclusively on other cybersecurity data sources, such as network traffic, malware binaries, or vulnerability datasets, are outside the scope unless security-related log data constitute a central component of the proposed approach. Second, this study does not develop, implement, or experimentally evaluate new AI models; instead, it synthesizes and critically analyzes existing literature. Finally, the conclusions of this review are inherently dependent on the quality, availability, and reporting practices of the selected publications.

1.8 Importance of the Study

With an explosion of Artificial Intelligence (AI) technologies in log analysis, there has emerged a growing body of literature that is both vast and highly fragmented. While there are many studies on using AI-based approaches for detection of anomalies, threats, root cause identification, and incident response, a variety of factors such as learning paradigms, security objectives, data, and evaluation methods used make the emergence of a coherent view of the field difficult. The importance of the proposed study is in providing a structured synthesis of the existing literature by creating a detailed taxonomy and comparative analysis. Through classification of the papers in accordance with AI paradigms, security objectives, and data from logs, the study proposes a framework that will help to better understand the existing literature and its methodology, strengths, and weaknesses, as well as research gaps. Moreover, by examining the evolution of AI techniques over time, the study provides insights into how security log analysis has progressed from traditional machine learning approaches toward transformer-based architectures, large language models, and intelligent autonomous security systems. Also, the findings are supposed to provide guidance to researchers, security analysts, and other professionals working in the area on choosing the appropriate AI methods for different cases of log analysis in the security domain and the design of intelligent security monitoring systems. Furthermore, by identifying open issues and promising areas for future research, such as explainable AI, transformer models, large language models, retrieval-augmented generation, root cause analysis, incident response, and self-healing security systems, this study sets the groundwork for further research into creating more reliable, adaptable, and autonomous AI-based cybersecurity solutions.

2 Research Methodology

2.1 Research Design

A Systematic Literature Review (SLR) serves as the research design for this study, with the purpose of providing a thorough, transparent, and reproducible synthesis of existing scientific literature on AI-based security log analysis. Contrary to a traditional narrative review, which is often prone to selection bias in choosing the literature and lacks structure in presenting the findings, an SLR involves a pre-established set of methods used to select, evaluate, and synthesize relevant scientific literature. This approach helps reduce selection bias, improves the methodological transparency of the review, and makes it reproducible. An SLR is especially suitable for the current study because the progress in the field of AI-based security log analysis has become rapid and fragmented at the same time. The existing scientific literature is characterized by substantial heterogeneity in terms of AI paradigm, security objectives, type of log data analyzed, experimental dataset, and methodology used for assessment of the proposed models, making it difficult to achieve a holistic understanding of the current state of the art through conventional narrative reviews. A systematic review appears to be more adequate for consolidation of different types of evidence and comparing the existing scientific works. Therefore, the main purpose of the research is not creating or evaluating an AI model but rather conducting a critical analysis, organization, and synthesis of the existing knowledge. The selected research design focuses on the synthesis of evidence rather than development of algorithms. With the help of a systematic review process, the literature will be classified based on AI paradigms, security objectives and log data used, thus facilitating the development of taxonomy and comparative analysis of the existing approaches. Furthermore, the chosen research design will allow identifying methodological strengths and limitations of the existing methodology, research trends and research gaps in the area.

2.2 Review Protocol

This Systematic Literature Review (SLR) follows a structured review protocol to guarantee a thorough, transparent, and reproducible performance of the Systematic Literature Review. The current review protocol primarily follows the guidelines set forth by Kitchenham and Charters (2007) [15] , who provide an extensive methodology for conducting systematic reviews in software engineering and other computer science domains. It appears to be suitable for the current study as the domain of AI-based security log analysis involves both software engineering and cybersecurity. Following the methodology provided by Kitchenham, this review is organized into three consecutive stages: planning the review, conducting the review, and reporting the review findings of the review. The planning stage encompasses setting out the goals of the research, research questions, the

scope, and the review protocol; the conducting stage includes systematic searching for relevant studies, applying pre-defined inclusion and exclusion criteria, quality assessment of the studies, extracting and synthesizing the evidence; the reporting stage involves the presentation of findings in terms of taxonomy, comparative analysis, trends in research, gaps, and directions for future research. In order to enhance transparency and reproducibility in the study selection procedure, this research also utilizes the PRISMA 2020 (Preferred Reporting Items for Systematic Reviews and Meta-Analyses) reporting guideline suggested by Page et al. (2021) [16]. Specifically, the PRISMA 2020 flow diagram will be used to document the procedures of identification, screening, eligibility assessment, and inclusion of studies. Thus, the combination of Kitchenham's methodology and the PRISMA 2020 reporting guideline establishes a robust methodological foundation that ensures minimal bias, enhanced methodological consistency, and transparency of the review process.

2.3 Search Strategy

The search strategy was designed to identify all relevant studies relevant to AI-based security log analysis. of security log analysis based on artificial intelligence (AI). As prescribed by the review protocol introduced in the previous section, the literature search was conducted across multiple reputable scientific databases and other sources of information covering the fields of artificial intelligence, cybersecurity, software engineering, and computer science. The primary sources included IEEE Xplore, ACM Digital Library, SpringerLink, and Scopus. Additional relevant studies that appeared in other reputable scientific sources and conferences, including arXiv, were used for this review as long as they contributed to achieving the goals of the review. This multi-source search strategy allows finding all high-quality papers without risks of missing relevant studies. Publications relevant for the review were retrieved according to a systematic search strategy based on predetermined keywords and Boolean operators. The set of keywords was developed considering the topics discussed in the scope of the review including the security logs, AI-related methods, and cybersecurity applications. Representative keywords included security logs, log analysis, log anomaly detection, artificial intelligence, machine learning, deep learning, transformer, large language model, retrieval-augmented generation, threat detection, root cause analysis, incident response, self-healing. These keywords were used together with the Boolean operators AND/OR to find publications dedicated to the various aspects of security log analysis using AI-based security log analysis. In order to reflect the recent development of the topic, the search was performed for publications published between 2020 and 2026 – years characterized by the rapid development of transformer-based methods, large language models, and retrieval-augmented generation techniques for security log analysis. Only studies published in English were considered

from consideration. The collected publications were then subject to the screening process according to the predetermined criteria of inclusion/exclusion, quality assessment, and data extraction as explained in the next sections.The retrieved studies were subsequently subjected to a structured screening process based on predefined inclusion and exclusion criteria, as documented through the PRISMA 2020 flow diagram and described in the following sections.

2.4 Study Selection Criteria

In line with the developed search strategy, the retrieved articles were selected through an systematic study selection process in order to include in this Systematic Literature Review only those which are relevant and of high methodologically sound. Selection was conducted based on specific criteria of inclusion and exclusion defined before the start of the review in order to reduce the risk of selection bias and increase consistency and reproducibility of the review. A study was considered eligible if the topic of its investigation includes the use of AI techniques to analyze security logs or cybersecurity-related tasks which use the log data. Studies were included if they investigated investigating issues of machine learning, deep learning, transformer-based approaches, large language models (LLMs), retrieval-augmented generation (RAG) and any other AI techniques utilized for security logs analysis for purposes of anomaly detection, threat detection, root cause analysis, incident response, explainability, and self-healing. Only peer-reviewed journal articles, conference papers and other scholarly publications in English and published within the timeframe from 2020 to 2026 were regarded as eligible for this review. On the contrary, the studies which do not address the problem of security logs analysis, discuss the issue of cybersecurity without usage of logs, are focused solely on the investigation of some other types of data (network traffic or malware binaries) and do not provide enough methodological information were considered ineligible.

Table 1: Inclusion and Exclusion Criteria

Inclusion Criteria	**Exclusion Criteria**
Studies applying AI-based techniques to security log analysis.	Studies not related to security log analysis.
Studies using log data as a primary source of security analysis.	Studies addressing cybersecurity without using log data.
Studies covering ML, DL, transformer-based models, LLMs, RAG, or related AI approaches.	Studies focusing exclusively on non-log data such as network traffic, malware binaries, or vulnerability datasets.
Studies addressing anomaly detection, threat detection, RCA, incident response, explainability, or self-healing.	Studies lacking sufficient methodological details.
Peer-reviewed journal articles, conference papers, and scientifically recognized publications published in English between 2020 and 2026.	Duplicate studies, non-English publications, or publications outside the selected timeframe.

2.5 Quality Assessment

To ensure the reliability and credibility of the review findings, all selected studies meeting the established inclusion criteria were systematically assessed for their quality prior to the synthesis of results. The aim of such quality assessment was to analyze the scientific quality of the selected studies and ensure that the evidence collected in this process comes from methodologically sound studies. Quality assessment is also important because it allows reducing the influence of methodological bias caused by unreliable experimental designs, lack of proper methodology, or unfounded conclusions from the selected studies. In this review, the quality assessment was conducted using a predefined quality assessment checklist according to the objectives of this study and generally accepted methods for conducting a systematic literature review. This checklist analyzes some methodological aspects of each selected study, including the clarity of the research objectives, the appropriateness of the proposed methodology, the relevance of the datasets used, the appropriateness of the evaluation methodology, the validity of the review findings, and the discussion of the study limitations. was evaluated on a uniform basis that ensures objectivity and reproducibility of the quality assessments. The resulting quality scores were considered in the synthesis and interpretation of the findings.

Table 2: Quality Assessment Checklist

ID	Quality Assessment Question	Score
QA1	Are the research objectives and the addressed security problem clearly stated?	0 / 0.5 / 1
QA2	Is the proposed AI methodology sufficiently described and reproducible?	0 / 0.5 / 1
QA3	Are the security log datasets or log sources appropriate and adequately described?	0 / 0.5 / 1
QA4	Is the experimental evaluation conducted using appropriate datasets, baselines, and evaluation metrics?	0 / 0.5 / 1
QA5	Are the reported results clearly presented and supported by the experimental evidence?	0 / 0.5 / 1
QA6	Are the limitations, challenges, or threats to validity discussed?	0 / 0.5 / 1

Each criterion was scored using a three-level scale: 1 for fully satisfied, 0.5 for partially satisfied, and 0 for not satisfied. The maximum possible quality score for each study was 6. The resulting scores were used to support the interpretation of the review findings rather than to automatically exclude studies from the synthesis.

2.6 Data Extraction

Data from each study selected in the literature search stage were systematically extracted using a specific data extraction form created specifically for this literature review. The goal of the data extraction process was to gather consistent and relevant data, which would facilitate the further stages of taxonomy creation, comparative analysis, research trends identification, and research gaps synthesis. In order to ensure consistency and avoid potential data extraction bias, the same criteria for data extraction were used for all the selected studies. Data extracted from each paper included bibliographical data (year of publication and publication outlet), the specific AI technique used, the security objective the paper aimed at addressing, the security log/log source type examined, the datasets used for validation of the findings, the experimental methodology and performance measures, the key findings of the study, its limitations, and future research directions. Gathering of such features allowed for comparing the selected studies and creating the basis for the analytical framework presented in the following chapters.

Table 3: Data Extraction Form

Data Item	Description
Study ID	Unique identifier assigned to each selected study.
Publication Year	Year in which the study was published.
Publication Venue	Journal, conference, repository, or scientific source where the study was published.
AI Technique	The AI method used, such as ML, DL, transformer-based models, LLMs, RAG, or related approaches.
Security Objective	The main security task addressed, such as anomaly detection, threat detection, root cause analysis, incident response, explainability, or self-healing.
Log Source	Type of log data analyzed, such as application logs, authentication logs, API logs, audit logs, system logs, or cloud/distributed logs.
Dataset	Dataset or log collection used for evaluation or validation.
Evaluation Metrics	Metrics used to assess the approach, such as accuracy, precision, recall, F1-score, AUC, or related measures.
Key Contribution	Main contribution or finding reported by the study.
Limitations	Limitations, challenges, or threats to validity reported by the authors.
Future Directions	Future research directions suggested by the study.

2.7 Data Synthesis

The extracted data were synthesized through a systematic qualitative synthesis in order to provide an comprehensive understanding of the current state of AI-based security log analysis research. Instead of merely providing a summary of each study, the synthesis identified the commonalities, methodologies, trends, and gaps in the selected research. Initially, the selected studies were classified on the basis of three primary dimensions: the AI methods used, security goals, and types of security log data analyzed. Such multi-criteria classification formed the basis for developing a comprehensive taxonomy of AI-based security log analysis. In addition, a comparative analysis was conducted to find out the similarities and differences in the selected papers in terms of methodologies, datasets, evaluation metrics, results, strengths, and limitations. Furthermore, a temporal analysis was carried out to examine the trends of the development of the area, showing how the research shifted from traditional machine learning to deep learning, transformer architecture, large language models, and retrieval-augmented generation. Finally, the

synthesized information was utilized to identify unresolved research gaps and emerging research opportunities and to highlight some research opportunities.

3 Literature Analysis and Taxonomy

3.1 Overview of Selected Studies

This chapter presents a critical synthesis of research on Artificial Intelligence (AI) techniques for security log analysis in distributed, cloud-based, and related computing environments. Its analytical scope follows the review design established in Chapter 2: the primary evidence base comprises original, peer-reviewed studies published between 2020 and 2026 that apply an AI-based method to log-derived information and evaluate an explicitly cybersecurity-related task. The chapter examines not only the architectures employed in the literature but also the security objectives pursued, the provenance of the analyzed records, the quality of experimental validation, and the limits of cross-study comparison.

The distinction between *security-specific evaluation* and *general log anomaly detection* is fundamental to the synthesis. Operational anomalies such as software errors, resource exhaustion, and ordinary system failures can reveal useful methods for parsing, representation learning, or event correlation. Nevertheless, improved detection of such anomalies does not, without further validation, demonstrate improved detection of malicious behavior. Accordingly, this review uses eligible studies to support the principal cybersecurity findings and refers to related operational or otherwise ineligible publications only where they provide relevant methodological context. This separation reduces the risk of transferring performance claims from unrelated tasks into the security domain.

3.1.1 Status of the Candidate Study Corpus

The working corpus comprises 28 individually numbered candidate studies, identified internally as R01–R28. At the time of drafting this chapter, 25 candidates had been examined, but eligibility decisions had been completed for only 24 of them. Five of these studies (R04 [17], R12 [1], R19 [8], R20 [18], and R22 [14]) were considered eligible or provisionally eligible for the primary synthesis, while 19 were retained as supporting literature or judged unsuitable for direct primary synthesis. The decision for R25 [5] remains unresolved because the paper does not sufficiently establish whether its GPT-based analysis uses original security log records or text generated from network-flow data. The remaining three candidates (R26–R28 [10, 19, 20]) have not yet completed screening. The six-question methodological quality assessment described in Chapter 2 (QA1–QA6) has also not been completed for the candidate primary set.

Table 4 records this interim evidence-selection status. These figures characterize the *working set of 28 candidate papers supplied for detailed review*; they are not totals of records retrieved, deduplicated, screened, or excluded in the complete database-search process. Consequently, they must not be presented as a completed PRISMA 2020 flow diagram or as final inclusion statistics.

Table 4: Interim eligibility status of the 28-study working corpus

Screening status	**Number**	**Interpretation**
Primary or provisionally eligible	5	R04 [17], R12 [1], R19 [8], R20 [18], and R22 [14]; quality appraisal pending.
Supporting literature or not eligible for primary synthesis	19	Background or methodological relevance without satisfying all primary inclusion requirements.
Eligibility decision pending	1	R25 [5]: original security-log provenance remains unverified.
Not yet screened	3	R26 [19], R27 [20], and R28 [10].
Total candidate papers	**28**	**Working corpus only; not PRISMA identification totals.**

The final number of included studies therefore remains undetermined. Likewise, a defensible PRISMA flow requires the original search records, database-specific retrieval totals, duplicate-removal counts, screening decisions, and reasons for full-text exclusion. Those quantities cannot be reconstructed reliably from the 28-study working corpus alone. Once screening is complete, the results reported in this chapter should be reconciled with the final study-selection log and the review protocol, including the requirement that primary publications have undergone peer review.

3.1.2 Characteristics of the Provisional Primary Evidence

The five currently eligible or provisionally eligible studies illustrate different combinations of AI technique, security objective, and log-source characteristics. They should not be treated as interchangeable implementations of a common detection task. Rather, they collectively demonstrate the heterogeneity that motivates the multidimensional taxonomy developed in Section 3.2.

R04 [17] investigates attack-oriented detection based on host audit and provenance information, using graph-based learning and knowledge distillation. Its security-focused evaluation supplies evidence directly related to malicious activity, while its graph representation emphasizes connections among system entities rather than isolated log messages. R12 [1] examines an integrated monitoring pipeline incorporating log collection,

LLM-assisted interpretation, retrieval-supported analysis, alerts, and workflow-oriented components. OpenSSH events provide a security-relevant use case; however, the study's largely qualitative evaluation constrains claims about quantitative improvements in detection or response effectiveness.

R19 [8] evaluates attention-enhanced recurrent learning for application-log anomalies, including user-related events with security implications as well as operational categories. Its reported aggregate classification results should therefore be interpreted in relation to mixed labels and should not be assumed to measure independently verified attack detection. R20 [18] examines sign-in behavior using an attention-enhanced LSTM autoencoder on Microsoft Entra ID authentication records. Its experimental labels identify behavioral irregularities rather than confirmed malicious logins, so its results are most directly relevant to authentication anomaly detection. R22 [14] compares syntactic feature extraction with Transformer-derived semantic representations using unsupervised learning on web, HTTP, and application-oriented datasets, including benchmarks containing attack-related requests. Its results are dependent on the evaluated dataset and algorithm, and its experimental design includes model-configuration choices informed by labels.

The candidate studies thus span relational audit evidence, authentication records, application-generated messages, and HTTP request data. They also represent different analytical functions: attack-oriented detection, anomaly classification, behavioral reconstruction, and analyst-facing interpretation. These differences affect the meaning of evaluation outcomes. For example, a weighted F1-score on a web-request benchmark, an anomaly-class F1-score on sign-in records, and a mixed-class aggregate F1-score on application logs are not measurements of the same underlying cybersecurity capability. Similarly, qualitative evidence that an LLM can summarize or contextualize an event does not establish that its explanations are factually grounded or that its recommendations improve incident response.

The provisional set also covers only part of the review's intended 2020–2026 publication window. Publication-year coverage within a small and unfinished candidate set should not be used to infer field-wide growth rates, publication shares, or the dominance of particular model families. More reliable temporal interpretation requires the final eligible corpus and consistent year-of-publication information, as discussed in Section 3.4.

3.1.3 Analytical Organization of the Chapter

The synthesis proceeds through complementary analytical stages. Section 3.2 develops a taxonomy based on AI techniques, security analysis objectives, and log-source types. Section 3.3 compares methodological choices, validation practices, reported outcomes, and practical constraints. Section 3.4 examines the temporal development of approaches represented in the reviewed literature, while Section 3.5 identifies recurring qualitative

research themes without interpreting them as statistically established trends. Finally, Section 3.6 derives an evidence-linked agenda of unresolved methodological and operational challenges.

This organization addresses the research questions defined in Chapter 2 by moving from characterization of the evidence base to structured classification, comparative evaluation, temporal interpretation, and identification of research gaps. Throughout these stages, *primary evidence* is distinguished from *supporting methodological literature*; reported performance is interpreted only in relation to the corresponding security task and dataset; and the conclusions remain explicitly provisional pending completion of the final screening and quality-assessment steps.

3.2 Taxonomy of AI-Based Security Log Analysis

The application of Artificial Intelligence (AI) to security log analysis encompasses a diverse range of computational approaches, security objectives, and data sources. This diversity creates challenges when attempting to systematically compare existing studies, particularly because different approaches may employ similar AI techniques while addressing substantially different security problems. Therefore, a structured taxonomy is necessary to organize the research landscape and establish meaningful relationships among existing methodologies.

Based on the analytical framework defined in Chapter 2, this review adopts a multidimensional taxonomy consisting of three principal dimensions: **AI techniques, security analysis objectives, and log data sources**. These dimensions represent complementary aspects of AI-based security log analysis and collectively provide a foundation for understanding the methodological characteristics and practical applicability of the reviewed approaches.

The first dimension, AI techniques, identifies the computational methods used to extract information, learn behavioral patterns, detect anomalous activities, and interpret security events. It encompasses traditional Machine Learning (ML), Deep Learning (DL), Transformer-based representation learning, Large Language Models (LLMs), and hybrid architectures incorporating Retrieval-Augmented Generation (RAG).

The second dimension, security analysis objectives, characterizes the cybersecurity problems addressed by the investigated approaches. These include intrusion detection, authentication anomaly detection, suspicious application activity identification, threat hunting, security incident analysis, and attack reconstruction. Such distinctions are important because general anomaly detection and confirmed cybersecurity threat detection do not necessarily represent equivalent evaluation tasks.

The third dimension, log data sources, distinguishes the information used as input to AI-based analysis. Relevant sources include application logs, authentication logs, host

audit logs, web access logs, and other security-related event records. The characteristics of these data sources influence preprocessing requirements, representation strategies, model selection, and the interpretation of detection outcomes.

Importantly, the proposed taxonomy is not intended to impose mutually exclusive categories on all studies. Hybrid approaches may incorporate multiple AI techniques, while individual studies may address more than one security objective or process several types of logs. Accordingly, the taxonomy distinguishes between the primary analytical function of an approach and the supporting techniques employed within its processing pipeline.

The following subsections examine these dimensions systematically, beginning with a classification of the AI techniques employed in security log analysis.

3.2.1 Classification Based on AI Techniques

AI-based security log analysis approaches can be classified according to the computational mechanisms through which they represent log events, learn behavioral patterns, and generate analytical outputs. Within the current review, five methodological categories provide a useful basis for organizing the approaches: traditional machine learning, deep learning, Transformer-based representation learning, large language models, and hybrid or retrieval-augmented architectures.

Traditional Machine Learning Approaches. Traditional ML techniques commonly operate on numerical features extracted from structured or semi-structured log records. These approaches may use statistical, syntactic, or semantically derived features to identify deviations from normal behavior. Clustering and anomaly detection algorithms, such as K-Means, Isolation Forest, and Self-Organizing Maps, can be applied without requiring labeled examples during the main model-training stage. Their effectiveness, however, depends substantially on the quality of the extracted features, the characteristics of the underlying logs, and the procedures used to configure and evaluate the models.

Karlsen et al. [14] investigated the relative effectiveness of syntactic and semantic feature extraction in web and application log anomaly detection. Their comparative evaluation demonstrated that semantic representations do not consistently outperform syntactic features across different datasets and learning algorithms. The study also highlighted the higher computational cost associated with Transformer-based embedding extraction. These findings indicate that the choice of feature representation should be considered in relation to both detection performance and computational efficiency.

Deep Learning Approaches. Deep learning methods employ neural architectures to learn complex relationships among log events and their associated features. Recurrent architectures, particularly Long Short-Term Memory (LSTM) networks, are relevant when the ordering or temporal characteristics of log events contribute to the analytical task.

Attention mechanisms may additionally support the selective weighting of sequence elements, although such weights should not automatically be interpreted as validated causal explanations.

Nusantara et al. [8] evaluated an attention-enhanced LSTM architecture for application log anomaly classification. The study incorporated security-relevant user anomalies alongside operational anomalies and reported improved aggregate classification performance compared with conventional recurrent baselines. Nevertheless, its rule-derived labels and limited attack-specific validation restrict the interpretation of these results as evidence of independently confirmed intrusion detection.

Similarly, Hizkia and Yugopuspito [18] investigated an attention-enhanced LSTM autoencoder using Microsoft Entra ID sign-in logs. Their approach modeled authentication behavior through sequence reconstruction and identified anomalous sequences based on reconstruction errors. The reported improvement over the baseline was modest, and the use of rule-derived anomaly labels emphasizes the importance of distinguishing behavioral deviations from verified malicious authentication events.

Graph-based deep learning constitutes another important methodological direction. In the study represented by R04 [17], graph-based models were applied to host audit data for Advanced Persistent Threat (APT) detection and attack reconstruction. The approach demonstrates how relationships among processes, files, and other system entities can be represented to support the identification of malicious activity extending beyond isolated log entries.

Transformer-Based Representation Learning. Transformer-based methods provide an alternative to conventional syntactic representations by encoding contextual relationships within log messages. Such approaches can support anomaly detection through pretrained language representations, task-specific adaptation, or embedding extraction. However, the use of a Transformer encoder for representation learning must be distinguished from the use of a generative language model for reasoning or response generation.

In R22 [14], DistilRoBERTa was used to generate semantic embeddings that were subsequently processed by conventional unsupervised learning algorithms. Thus, the Transformer served as a feature extractor rather than an end-to-end security decision-making system. This architectural distinction is relevant when comparing model complexity, computational cost, and the source of improvements in classification performance.

Large Language Model-Based Approaches. Large Language Models extend log analysis beyond numerical classification by supporting natural-language interpretation, contextual summarization, and the generation of diagnostic explanations. In security applications, these capabilities may help analysts interpret suspicious behavior and formulate investigation recommendations. Nevertheless, generated explanations are not necessarily factually reliable, and their security relevance must be established through appropriate evaluation.

Pundlik et al. [1] incorporated LLM-based analysis into an integrated log-monitoring framework that included OpenSSH security events, alert generation, and automated operational workflows. The study illustrated the feasibility of integrating language models into log-monitoring pipelines, but its predominantly qualitative evaluation limits conclusions regarding measurable improvements in threat-detection accuracy.

Hybrid and Retrieval-Augmented Architectures. Hybrid architectures combine complementary analytical components to address limitations associated with individual models. These combinations may include neural anomaly detectors, graph-based representations, language models, knowledge retrieval, and rule-based validation. Retrieval-Augmented Generation is particularly relevant when a language model requires contextual information from historical logs or other external evidence.

The framework examined in R12 [1] represents an example of retrieval-supported log analysis, in which contextual information is retrieved and supplied to language models to assist interpretation. Such architectures offer potential advantages for contextualizing unfamiliar events, but their effectiveness depends on retrieval relevance, the reliability of generated interpretations, and the availability of security-specific validation.

Table 5 summarizes the principal AI techniques represented in the studies provisionally eligible for the primary synthesis. The categories describe methodological roles rather than mutually exclusive model families.

Table 5: AI technique classification of provisionally eligible primary studies

Study ID	**Primary AI approach**	**Supporting technique**	**Main security application**
R04 [17]	Graph-based deep learning	Knowledge distillation	APT detection and attack reconstruction
R12 [1]	LLM-based analysis	RAG and log monitoring integration	Authentication threat analysis and alerting
R19 [8]	Recurrent deep learning	Attention mechanism	Application log anomaly detection
R20 [18]	LSTM autoencoder	Attention and reconstruction-error scoring	Authentication anomaly detection
R22 [14]	Unsupervised machine learning	TF-IDF and DistilRoBERTa embeddings	Web and application request anomaly detection

The classification reveals that the reviewed approaches differ not only in architectural complexity but also in the analytical functions assigned to their AI components. Deep learning methods commonly provide behavioral modeling and anomaly classification, whereas graph-based methods emphasize relationships among entities and events. Transformer representations facilitate semantic feature extraction, while LLM-based and

retrieval-augmented methods extend the processing pipeline toward contextual interpretation and analyst support.

However, increasing architectural sophistication does not establish superior security effectiveness. The reviewed studies differ substantially in their datasets, labeling procedures, security tasks, and evaluation metrics. Consequently, their reported performance values should not be compared as though they were obtained under equivalent experimental conditions. A more meaningful assessment requires consideration of both the security objective and the characteristics of the underlying log sources, which are examined in the following subsections. """

3.2.2 Classification Based on Security Analysis Objectives

In addition to the computational techniques used, the reviewed approaches can be differentiated by the security objectives they seek to accomplish. This distinction is necessary because the term *log anomaly detection* covers substantially different analytical outcomes. An unusual event may represent a legitimate change in system behavior, an operational failure, a policy violation, or malicious activity. Consequently, evidence that a model distinguishes normal from anomalous log records cannot, by itself, establish its effectiveness in identifying cybersecurity attacks. A security-objective-based classification therefore considers the intended analytical output, the relationship between the evaluated data and a cybersecurity threat, and the strength of the corresponding validation.

Within the provisionally eligible primary studies, four overlapping application areas can be identified: (i) attack-oriented threat detection and reconstruction, (ii) authentication anomaly detection, (iii) suspicious web and application activity detection, and (iv) security-event interpretation and alerting. These areas represent the objectives supported by the available evidence rather than an exhaustive inventory of all potential applications of AI to security operations.

Attack-Oriented Threat Detection and Reconstruction. Attack-oriented methods aim to identify behaviors associated with malicious actions, rather than merely deviations from a statistical baseline. In host environments, suspicious activities may involve relationships among processes, files, and other entities that are difficult to recognize when log entries are evaluated independently. Graph-based representations are therefore relevant because they preserve relationships among events and the entities involved.

The approach examined in R04 [17] applies graph-based learning and knowledge distillation to host audit information for security threat detection. By representing interactions among system entities, it supports analysis of attack-related behavioral context beyond isolated events. Its evaluation on security-oriented audit and provenance datasets provides more direct evidence of attack detection than evaluations based exclusively on

generic operational faults. Nevertheless, threat detection and complete incident reconstruction remain distinct outcomes; evidence for detecting malicious behavior should not automatically be interpreted as proof that an entire attack sequence can be reconstructed accurately.

Authentication Anomaly Detection. Authentication-focused approaches analyze sign-in activity to identify behavior that differs from established user or organizational patterns. Such analysis may draw on event timestamps, source IP addresses, geographic locations, device characteristics, authentication outcomes, and multifactor authentication (MFA) results. These features are particularly relevant to identifying potentially suspicious access attempts, although unusual authentication behavior is not inherently malicious.

R20 [18] investigates this objective using Microsoft Entra ID interactive sign-in logs and an attention-enhanced LSTM autoencoder. The model learns sequential login patterns and uses reconstruction error to identify anomalous sequences. The reported anomaly-class F1-score was 0.81, compared with 0.80 for the baseline LSTM autoencoder. However, the reference anomaly labels were derived from behavioral rules, including unusual locations and MFA-related conditions. Accordingly, the reported performance measures agreement with those anomaly labels; it does not establish the detection rate for independently confirmed account compromise or credential-based attacks. This distinction is important when interpreting authentication anomaly detection as a security capability rather than as a general behavioral deviation detector.

Suspicious Web and Application Activity Detection. Web and application log analysis seeks to identify suspicious requests, potentially malicious interactions, and deviations in application-level behavior. Unlike host audit analysis, the relevant evidence may be contained in HTTP request records, access logs, and application-generated event messages. The security implications depend on both the provenance of the records and whether the evaluation includes documented attack behavior.

R22 [14] evaluates unsupervised anomaly detection using syntactic and semantic representations of application and HTTP data. Two of its datasets, ECML/PKDD 2007 and HTTP CSIC 2010, contain examples of attack-related web requests, allowing the experimental results to be interpreted within a web-security context. However, these datasets include simulated or benchmark attack scenarios and do not, by themselves, demonstrate effectiveness against previously unseen attacks in operational deployments. In addition, the reported metric is a weighted F1-score, which should not be treated as directly comparable to an anomaly-class F1-score obtained in another study.

R19 [8] also addresses application log anomaly detection using an attention-enhanced LSTM architecture. Its evaluation includes user-related anomalies associated with sus-

picious authentication or unauthorized activity alongside system-related anomalies. Although these user events provide security relevance, the labeling strategy relies on predefined rules, and the reported aggregate classification metrics do not isolate performance against independently verified attack events. R19 therefore illustrates a broader application-monitoring task in which security-relevant and operational anomalies coexist.

Security-Event Interpretation and Alerting. Security-event interpretation aims to support analysts in understanding potentially harmful activity and determining which events require investigation. In contrast to classification-only approaches, interpretation-oriented systems may produce natural-language explanations, contextual summaries, or alerts that can facilitate human decision-making. These outputs, however, must be distinguished from validated automated incident response, which would require evidence of incident containment, remediation, or other operational actions.

R12 [1] examines an integrated monitoring approach that combines log collection, LLM-assisted analysis, alerting, and retrieval-supported components. Its inclusion of OpenSSH-related security events provides a direct authentication-security use case. At the same time, the study's predominantly qualitative evaluation does not establish a reliable quantitative improvement in detection precision, recall, or incident-response effectiveness. The contribution is therefore best characterized as the integration of security log monitoring and analyst-oriented interpretation, rather than as demonstrated autonomous response to cybersecurity incidents.

Cross-Cutting Objectives and Evidence Boundaries. Some security objectives are closely related but should not be conflated. *Anomaly detection* identifies behavior that differs from an expected pattern; *threat detection* requires a defensible relationship between the detected behavior and malicious activity; *incident analysis* attempts to explain or contextualize security events; and *incident response* involves decisions or actions taken to investigate, contain, or remediate an incident. An approach may contribute to several of these functions without providing equally strong evidence for all of them.

This distinction is especially relevant when evaluating explainability claims. Attention weights, provenance relationships, and generated summaries can provide potentially useful investigative context, but they are not interchangeable forms of explanation and do not necessarily establish causal correctness. Similarly, the presence of alerting or self-healing components in a proposed architecture should not be interpreted as demonstrated autonomous security remediation unless the associated actions are evaluated under explicit security scenarios.

Table 6 maps the provisionally eligible studies to their principal security objectives and indicates the main boundary of the supporting evidence. Multiple studies may address related objectives, but each is assigned the primary role most clearly supported by its

experimental design.

Table 6: Security analysis objectives and evidence boundaries in provisionally eligible primary studies

Study	**Primary security objective**	**Evaluated log or event source**	**Main evidence boundary**
R04 [17]	Attack-oriented threat detection	Host audit and provenance records	Attack detection evidence does not automatically establish complete attack reconstruction.
R12 [1]	Security-event monitoring and alerting	OpenSSH and integrated monitoring logs	Limited quantitative validation of threat detection and response effectiveness.
R19 [8]	Security-relevant application anomaly detection	Application logs with user and system anomalies	Rule-derived labels and aggregated results limit attack-specific inference.
R20 [18]	Authentication anomaly detection	Microsoft Entra ID sign-in logs	Behavioral anomaly labels are not independently verified account-compromise incidents.
R22 [14]	Web and application attack-related anomaly detection	HTTP request data and web access logs	Benchmark or simulated attacks and label-informed configuration limit generalization.

The resulting classification suggests that the currently eligible evidence is concentrated on identifying suspicious or anomalous activity, with fewer directly evaluated capabilities for explaining incidents and supporting downstream security operations. Importantly, this observation concerns the provisionally eligible set reviewed at this stage; it should not be generalized into a quantitative conclusion about the entire research field before study selection is finalized.

The security-objective dimension also reveals a methodological challenge for comparative synthesis. Studies that report accuracy or F1-scores for rule-derived anomalies, simulated web attacks, and labeled attack provenance graphs are not necessarily measuring the same security outcome. Meaningful comparison therefore requires separating detection targets, label provenance, dataset realism, and evaluation metrics. The log data sources underlying these distinctions are considered in Section 3.2.2, which follows.
"""

subsubsectionClassification Based on Security Log Data Sources

The type and provenance of log data constitute a third essential dimension of the proposed taxonomy. Although many AI-based approaches share similar modeling architectures, their security relevance and practical requirements can differ substantially according to the events recorded, the entities represented, and the operational context in which the records are generated. For instance, sign-in events provide evidence of authentication behavior, whereas host audit records capture interactions among operating-system enti-

ties. HTTP request records reflect application-facing activity, while application-generated messages may combine security-relevant events with ordinary errors and diagnostic information. Consequently, a classification based solely on the AI technique or reported evaluation metric would obscure differences in the underlying security evidence.

This review distinguishes log sources according to their original event-producing systems and analytical purpose rather than their textual appearance after preprocessing. A structured network-flow record does not become an authentication or audit log merely because its numerical fields have been converted into a natural-language description. Similarly, an operational error message does not provide evidence of a malicious incident unless the evaluation establishes a defensible security relationship. These distinctions are particularly important for studies that transform network telemetry into text before processing it with language models, or that evaluate log-based anomaly detection using datasets dominated by system faults.

Host Audit and Provenance Logs. Host audit logs document interactions among operating-system entities, including processes, files, and other resources. Their value for security analysis lies not only in individual events but also in the relationships through which suspicious actions propagate. Provenance representations can connect activities over time and support the analysis of attack-related behavior that may be difficult to identify from isolated log messages.

R04 [17] applies graph-based techniques and knowledge distillation to audit-derived provenance information in the context of threat detection. This application illustrates why relational structure is an important characteristic of host-level security evidence: modeling the connections among processes and resources can preserve context that would be lost if each log entry were treated as independent. However, the availability of provenance relationships, their granularity, and the security labels used during evaluation must be considered before comparing such results with those obtained from conventional text logs. Host audit data may also impose substantial collection and processing demands that are not represented by classification metrics alone.

Authentication and Identity-Related Logs. Authentication logs record attempts to establish or verify user identities and may include timestamps, source addresses, locations, client details, success or failure outcomes, and multifactor authentication information. These features make such logs suitable for behavioral modeling and for identifying potentially suspicious access patterns. Nevertheless, an unusual login may reflect legitimate travel, device changes, or administrative activity rather than a confirmed attack.

R20 [18] uses Microsoft Entra ID interactive sign-in records collected from an organizational setting. The study examines temporal login patterns through an attention-enhanced LSTM autoencoder and evaluates anomalous sequences using rule-derived la-

bels. Its design therefore provides direct evidence of authentication-behavior anomaly detection, but it does not establish that every labeled anomaly corresponds to a malicious login. R12 [1] also includes OpenSSH-related events within a broader monitoring and alerting workflow, illustrating how authentication-oriented records can serve as inputs to LLM-assisted interpretation. Because the evaluation strategies of R12 and R20 differ, their outputs should not be combined into a single measure of authentication-threat detection effectiveness.

Application-Generated Security Events. Application logs originate from software components and may describe user interactions, authorization outcomes, application errors, and other events generated during execution. These sources are heterogeneous: an individual dataset may contain records relevant to both security monitoring and operational reliability. Their utility for cybersecurity analysis therefore depends on the particular events selected, the semantics of the associated fields, and the process used to establish security-related labels.

R19 [8] analyzes application logs with an attention-enhanced LSTM architecture and considers both user-related and system-related anomaly categories. This mixed event population is methodologically important. High aggregate classification performance does not necessarily imply equally strong performance on the security-relevant subset, particularly when labels are derived from predefined rules rather than independent incident investigations. Separating user-security events from general system anomalies is therefore necessary when assessing what the study demonstrates about cybersecurity-specific detection.

Web Access and HTTP Request Records. Web access logs and HTTP request records describe interactions with application endpoints and may contain request methods, paths, query strings, status information, and other request-level attributes. Such records can support the identification of suspicious web activity when anomalous or malicious requests are represented and labeled in the evaluation dataset. However, the interpretation of these records depends on the distinction between actual application access events, benchmark request datasets, and network-flow summaries.

R22 [14] evaluates syntactic and semantic feature extraction using HTTP and application-oriented datasets, including ECML/PKDD 2007 and HTTP CSIC 2010, which contain examples of normal and attack-related requests. These datasets provide a direct web-security evaluation context, although simulated attacks and benchmark-specific data characteristics constrain the generalizability of the findings. In particular, the reported weighted F1-scores reflect the datasets and class distributions used in the study; they cannot be directly equated with anomaly-class F1-scores reported on organizational authentication logs.

Cloud, Distributed-System, and Aggregated Security Logs. Cloud and distributed environments commonly produce records across applications, identity services, hosts, containers, and network components. Such records differ in event granularity, timestamp accuracy, formatting, and the availability of cross-system identifiers. These properties create practical challenges for correlation and interpretation, especially when a security incident spans multiple components. For analytical purposes, however, a multi-source collection should be categorized by the security-relevant event types it actually contains rather than by the deployment environment alone.

The provisional primary evidence includes integrated monitoring in R12 [1] and application-level records in R19 [8], but it does not yet establish a sufficiently broad basis for treating all categories of cloud and distributed telemetry as independently evaluated security-log sources. Studies concerned with operational faults, log–metric correlation, or generic system anomalies may inform the engineering of such pipelines; under the review protocol, they remain supporting literature unless their evaluation directly addresses a cybersecurity-specific task. This distinction preserves the boundary between security-log analysis and general AIOps.

Dataset Provenance, Ground Truth, and Comparability. Across these log-source families, the quality of the evidence depends on more than the number of records or the sophistication of the model. Three considerations are especially important. First, *event provenance* concerns whether inputs originate from the stated logging system and retain the security context required for interpretation. Second, *label provenance* concerns how normal, anomalous, or malicious events were identified, whether by rule-based heuristics, simulated attack scenarios, existing benchmark labels, or independently verified incidents. Third, *evaluation granularity* concerns whether results are reported per log entry, event sequence, entity, or attack scenario. Differences along any of these dimensions can change what a performance metric actually measures.

These considerations are apparent in the provisionally included studies. R20 [18] evaluates sequences of authentication events against behavioral anomaly labels; R19 [8] combines security-related user anomalies with operational categories; R22 [14] evaluates web and HTTP data containing benchmark attack behavior; and R04 [17] analyzes host-level relational audit evidence. Even if two studies report the same F1-score, the results do not necessarily support equivalent conclusions about practical cybersecurity effectiveness. The use of different ground-truth procedures also complicates comparison between anomaly detection and verified threat detection.

Table 7 summarizes the principal log-source categories represented in the provisionally eligible primary literature. The table emphasizes the nature of the input evidence and the corresponding limitations of security interpretation rather than attempting to rank the sources by importance.

Table 7: Security log-source categories and their analytical characteristics

Log-source category	**Study examples**	**Security-relevant evidence**	**Interpretive limitation**
Host audit and provenance	R04 [17]	Relationships among processes, resources, and attack-related actions.	Detection results depend on provenance completeness and attack-ground-truth quality.
Authentication and identity	R20 [18]; R12 [1]	Login outcomes, source context, timing, MFA, and SSH authentication-related events.	Behavioral deviations are not automatically verified unauthorized access.
Application-generated events	R19 [8]	User-related anomaly indicators alongside application and system events.	Mixed operational and security labels may obscure attack-specific performance.
Web access and HTTP requests	R22 [14]	Access/request characteristics with benchmark normal and attack behavior.	Benchmark and simulated requests may not represent deployed application traffic.

The log-source dimension complements the AI-technique and security-objective dimensions by clarifying the evidence on which each system operates. In particular, it demonstrates why an apparently successful anomaly detector trained on system faults cannot be treated as equivalent to a detector evaluated on attack-labeled HTTP requests or host audit provenance. The taxonomy therefore requires explicit consideration of data origin, event semantics, labeling procedures, and the unit of evaluation when interpreting results.

Taken together, the three dimensions provide a consistent framework for subsequent comparative analysis. AI techniques characterize *how* log information is processed; security objectives specify *what* the system is intended to accomplish; and log data sources establish *which evidence* supports the reported findings. The comparative synthesis in Section 3.3 applies these distinctions to assess the methodological strengths, performance claims, and limitations of the reviewed approaches without assuming that heterogeneous datasets or evaluation metrics are directly interchangeable. """

3.3 Comparative Analysis of Existing Approaches

The taxonomy introduced in Section 3.2 distinguishes AI techniques, security objectives, and log data sources. Building on these dimensions, this section critically compares the methodological design and empirical evidence of the studies provisionally eligible for the primary synthesis. Rather than treating all reported performance results as directly interchangeable, the comparison considers how each study defines the target security task, represents log data, constructs its reference labels, selects evaluation baselines, and

supports its principal claims. Such an evidence-centered comparison is necessary because a high score on a particular log dataset does not necessarily imply equivalent performance in a different cybersecurity setting.

At the current stage of the review, the comparative set comprises R04 [17], R12 [1], R19 [8], R20 [18], and R22 [14]. Their eligibility remains provisional pending completion of screening and methodological quality assessment. R25 [5] is not included because the provenance of its reported log inputs remains unresolved, while R26–R28 [10, 19, 20] have not yet undergone eligibility assessment. Accordingly, the following findings describe the *current candidate set* and should not be interpreted as final estimates for the literature as a whole.

3.3.1 Comparison of Methodological Architectures

The candidate studies implement markedly different analytical pipelines. R04 [17] applies a teacher–student graph-learning approach to provenance representations derived from host audit events. Its design emphasizes relationships among system entities and uses knowledge distillation to obtain a lightweight detection model. This relational representation is suitable for attack-oriented analysis in which suspicious behavior is distributed across several processes or resources. However, graph construction and the availability of sufficiently complete provenance records constitute methodological dependencies that differ from those of conventional textual log classification.

R19 [8] employs a recurrent neural model with an attention mechanism to classify application log records. The approach transforms log messages into numerical sequences and learns distinctions among normal events, user anomalies, and system anomalies. Its principal advantage within the reported setting is supervised discrimination among these categories, although the resulting predictions inherit the meaning and limitations of the rule-based labeling process. R20 [18] instead uses an attention-enhanced LSTM autoencoder to reconstruct sign-in sequences and identify deviations from normal authentication patterns. In contrast to R19's classification-oriented formulation, R20's anomaly score depends on reconstruction error and a selected decision threshold.

R22 [14] examines a different architectural question: whether syntactic or semantic representations are more effective when supplied to unsupervised anomaly detection algorithms. It compares TF–IDF features with DistilRoBERTa embeddings across K-Means, Self-Organizing Maps (SOM), Isolation Forest, and agglomerative clustering. The resulting performance therefore reflects the interaction between the feature representation, the selected algorithm, and the dataset rather than the effectiveness of a single end-to-end neural detector. Notably, the use of class labels during parts of the hyperparameter-selection procedure complicates a strictly label-independent interpretation of the experimental protocol.

R12 [1] extends beyond detection-only architectures by integrating ELK-based monitoring, retrieval-supported LLM analysis, alerting, and operational automation. The implementation demonstrates how log collection, contextual interpretation, and workflow triggers can be connected in a single system. Unlike the other candidates, however, R12 does not present a comparable quantitative attack-detection benchmark. Its contribution is therefore assessed primarily in terms of integration feasibility and the documentation of example outcomes, rather than superior predictive performance.

Table 8 summarizes these differences at the level of analytical tasks, source evidence, and evaluation design.

Table 8: Methodological comparison of provisionally eligible primary studies

Study	**Modeling approach**	**Input data and task**	**Evaluation design**
R04 [17]	Graph-based teacher–student learning with knowledge distillation	Host audit provenance; threat detection and attack-trace analysis	Attack-related provenance benchmarks, including StreamSpot, Unicorn Wget, and DARPA E3 subsets; baseline comparisons and ablation experiments
R12 [1]	ELK monitoring, LLM-assisted analysis, RAG, and workflow automation	Application and OpenSSH logs; incident interpretation, alerting, and remediation workflows	Demonstration on a sample application and analysis of 2,000 historical OpenSSH entries; mainly qualitative assessments
R19 [8]	Attention-enhanced LSTM classification	Application logs labeled as normal, user-anomalous, or system-anomalous	Comparison with RNN, GRU, and standard LSTM; rule-derived labels and class balancing
R20 [18]	Attention-enhanced LSTM autoencoder with reconstruction-error scoring	6,314 Microsoft Entra ID sign-in records; authentication anomaly detection	Comparison against a non-attention autoencoder; behavioral heuristic labels and threshold selection
R22 [14]	TF–IDF versus DistilRoBERTa embeddings with four unsupervised algorithms	Apache access and HTTP request datasets; application/web anomaly detection	Three datasets, 75/25 stratified split, algorithm comparison, and computational-cost analysis

3.3.2 Comparison of Datasets and Validation Practices

The candidate studies use different sources of security evidence. R04 [17] relies on graph-structured provenance benchmarks containing attack-related activity, allowing the authors to evaluate a model designed for malicious entity or behavior identification. R20 [18]

uses organizational sign-in records in which anomalies are defined through behavioral criteria rather than independently confirmed account compromise. R19 [8] combines user-related anomalies with operational system anomalies, meaning that the aggregate task is broader than attack detection alone. R22 [14] includes web and HTTP request benchmarks with normal and attack-related requests, alongside an application access dataset. R12 [1] combines logs from a demonstration application with historical OpenSSH events, including authentication-related behavior.

These data sources differ along at least four dimensions relevant to validity. First, *security specificity* varies: some datasets include labeled attack scenarios, whereas others primarily capture operational deviations that may or may not be security-related. Second, *label provenance* varies between attack scenarios, benchmark labels, and rules or heuristics. Third, *unit of analysis* differs among audit entities or graphs, individual application messages, authentication sequences, and web requests. Fourth, *deployment realism* differs between controlled benchmarks, historical organizational records, and system demonstrations. Without accounting for these differences, a comparison based on F1-score alone would conflate distinct research questions.

The experimental procedures introduce further sources of uncertainty. In R19 [8], class balancing is performed on a heavily imbalanced application-log dataset, which changes the prevalence of the classes represented during evaluation. Consequently, the reported classification metrics should not be assumed to reflect the false-alarm burden of an unmodified production log stream. In R20 [18], the anomaly threshold is chosen to maximize F1 against reference labels, so the independence of threshold calibration from the final test assessment is a key consideration for methodological appraisal. In R22 [14], although the clustering algorithms are characterized as unsupervised, class labels inform certain choices of algorithm configuration; this may produce optimistic evaluation if configuration selection and final testing are not rigorously separated. For R12 [1], the limited availability of quantitative ground truth prevents a conventional estimate of detection precision or recall.

An important implication is that *absence of a directly comparable metric does not mean absence of a contribution.* R12 [1] provides architectural and workflow evidence that differs in kind from the predictive metrics in R19 [8] or R20 [18]. Likewise, R04 [17]'s graph-level threat evidence cannot be reduced to the same task as web request clustering in R22 [14]. The review therefore uses structured qualitative synthesis rather than calculating a pooled accuracy or constructing a numerical ranking across these heterogeneous studies.

3.3.3 Comparison of Reported Experimental Findings

The available numerical evidence supports meaningful *within-study* comparisons but not direct ranking across all candidates. In R19 [8], the authors report an overall accuracy of 98.52%, precision of 98.66%, recall of 98.52%, and F1-score of 98.55% for their attention-enhanced LSTM. These values pertain to a three-category application-log classification setting containing normal, user-anomaly, and system-anomaly labels. They should not be represented as verified intrusion-detection rates because the task includes operational anomalies and its labels were constructed using predefined rules.

For R20 [18], the anomaly-class precision, recall, and F1-score of the attention-enhanced LSTM autoencoder are 0.76, 0.87, and 0.81, respectively. The non-attention baseline reports an anomaly-class F1-score of 0.80. The observed improvement is therefore *0.01 in F1*, or one percentage point when expressed on a 0–100% scale. This modest gain supports a limited within-study benefit from attention but not a claim that attention is generally superior across authentication datasets. Furthermore, both models are evaluated against the study's behavioral anomaly labels, not a separately verified collection of account-takeover incidents.

R22 [14] reports weighted F1-scores for multiple combinations of representations, algorithms, and datasets. Its results demonstrate that the preferred representation depends on the dataset and model. The paper reports differences of approximately 2% favoring semantic features on Apache access, approximately 6% and 8% favoring semantic features for SOM and K-Means on ECML/PKDD, and approximately 10% favoring syntactic features for SOM on HTTP CSIC. The paper does not clearly establish whether these reported percentages represent relative changes or percentage-point differences. Because the paper presents the main F1 comparisons graphically rather than as a full numeric results table, these differences should be treated as the authors' approximate comparisons. The computational-cost table also reports substantially longer feature extraction for semantic embeddings than for syntactic features: 44 versus 5 seconds on Apache access, 138 versus 11 seconds on CSIC, and 217 versus 27 seconds on PKDD. Thus, representation quality and computational efficiency involve an explicit trade-off within the reported settings.

R04 [17] evaluates threat detection across several audit-provenance benchmarks and compares the distilled student against graph-based teacher models and other methods. The experiments report dataset- and configuration-dependent detection performance, together with analyses of efficiency and model components. For example, the available study version reports F1-scores between 97.33% and 99.69% for the student when using a GCN teacher across five evaluated datasets. This range is *descriptive of that particular reported configuration and set of benchmarks*, not a pooled measure of general threat-detection effectiveness or an estimate transferable to other log sources. In addition, the

existence of an attack reconstruction component does not mean that its quality is measured by the detection F1-score.

R12 [1] reports demonstrations of log categorization, explanation, alerting, and workflow triggers, as well as a qualitative comparison of three LLM configurations. The comparison distinguishes whether generated responses followed the requested schema and whether their content was assessed as appropriate. However, it does not provide a labeled detection benchmark with precision, recall, or F1-score, nor a quantitative assessment of security-remediation success. Accordingly, these observations should be characterized as feasibility evidence, not as a measured improvement in threat-detection accuracy.

Table 9 preserves the task and metric context needed to interpret the principal findings. A cross-study performance chart is deliberately omitted: plotting the reported figures on a shared axis would create a misleading impression of equivalence between different targets, label types, and metric definitions.

Table 9: Reported findings and limits of cross-study performance comparison

Study	**Reported result**	**Metric or evaluation target**	**Interpretation boundary**
R04 [17]	GCN-teacher student F1: 97.33–99.69% across five reported datasets	Dataset-specific threat detection on provenance benchmarks	No transferable pooled threat-detection score; does not quantify reconstruction quality
R12 [1]	Qualitative schema/content comparisons and illustrated incident-handling workflows	LLM output inspection and sample operational/security cases	No attack-specific precision, recall, F1, or measured remediation success
R19 [8]	Accuracy 98.52%; F1 98.55%	Overall three-class application-log classification	Combines security-relevant user anomalies with operational anomalies; rule-based labels
R20 [18]	Anomaly F1: 0.81 with attention versus 0.80 without; precision 0.76; recall 0.87	Authentication-sequence anomalies against behavioral labels	Not a confirmed malicious-login detection rate; threshold selection matters
R22 [14]	Semantic versus syntactic weighted F1 differences vary by dataset and algorithm; semantic extraction slower	Weighted F1 and processing time for unsupervised web/application anomaly analysis	No single comparable F1; benchmark dependence and label-informed tuning

3.3.4 Strengths, Limitations, and Practical Applicability

The principal methodological strength of R04 [17] is the preservation of relational context in host audit evidence, together with a student model designed to reduce computational demands. Its practical applicability depends on the availability and completeness of host provenance records, and its published benchmark results require validation under different deployment conditions and threat behaviors. R19 [8] benefits from an explicit application-log processing pipeline and comparisons with recurrent baselines, but its reliance on rule-derived classes and combined operational/security outcomes limits attack-specific inference. R20 [18] focuses on a concrete authentication log source and offers a direct baseline comparison, yet the small F1 improvement and heuristic reference labels constrain the conclusions that can be drawn about confirmed account compromise.

R22 [14] provides an especially useful representation-level comparison because it examines several learning algorithms and explicitly considers computation time. The results challenge the assumption that a more complex semantic encoder always provides a better or more efficient anomaly detector. However, the dependence on benchmark request data and label-informed parameter selection limits generalizability and complicates the interpretation of the learning procedure as wholly unsupervised. R12 [1] contributes a broader systems perspective through integrated monitoring, contextual analysis, alerting, and automated workflows. At present, its strongest evidence concerns the feasibility of connecting these components rather than the effectiveness or safety of autonomous security response under independently verified incidents.

For deployment decisions, predictive performance is therefore only one dimension of suitability. Data collection cost, model complexity, latency, data confidentiality, resilience to evolving log formats, availability of reliable labels, and the consequences of false positives or false negatives may be equally important. The candidate literature does not provide sufficiently standardized measurements of all these properties to support a definitive ranking of algorithms. In particular, automatic actions driven by model outputs introduce operational risks that cannot be evaluated solely through examples of correctly generated explanations or triggered workflows.

3.3.5 Cross-Study Synthesis and Implications for Research Question RQ3

The comparative evidence suggests three provisional conclusions. First, *the relationship between log representation and security objective is central to method selection.* Graph-based representations are suited to relational host activity, sequence models address temporal patterns in authentication or application logs, and syntactic or semantic embeddings can support web request anomaly analysis. The suitability of a representation must be established against the corresponding security task rather than inferred from general architectural sophistication.

Second, *evaluation validity depends strongly on label provenance and data realism.* Rule-defined anomalies, controlled attack benchmarks, and independently verified malicious incidents provide different levels and types of security evidence. Reported performance should therefore be interpreted alongside the source of ground truth, the unit of analysis, the class distribution, and any label-dependent tuning decisions.

Third, *interpretation and response capabilities are less consistently quantified than detection outcomes within the current candidate set.* R12 [1] demonstrates a pipeline for LLM-assisted analysis and workflow execution, but does not provide metrics equivalent to those used for supervised or anomaly-based detection. This difference identifies an important limitation in the available evidence rather than demonstrating that one architectural category is intrinsically better or worse.

These findings address RQ3 by showing that existing AI approaches cannot be compared adequately through headline accuracy or F1-score alone. Their methodological strengths and limitations emerge from the interaction among security objectives, log-source properties, model design, and experimental validation. The analysis remains provisional until the screening of remaining papers, resolution of the R25 [5] data-provenance question, and application of the QA1–QA6 checklist have been completed. The next section examines the temporal development of the research landscape without assuming that the availability of newer AI techniques necessarily implies improved cybersecurity effectiveness.

3.4 Evolution of AI-Based Security Log Analysis

The development of AI-based security log analysis can be examined through changes in log representation, learning architectures, and the functions assigned to intelligent components within security-monitoring pipelines. However, the appearance of a new modeling paradigm does not imply that earlier approaches have become obsolete or that the newer approach necessarily produces better cybersecurity outcomes. The reviewed studies instead suggest an increasingly diverse methodological landscape in which syntactic features, sequence modeling, relational representations, semantic embeddings, and language-model-assisted interpretation are used for different analytical purposes.

This section addresses Research Question RQ4 by examining these developments within the publication period specified in Chapter 2 (2020–2026). The discussion distinguishes between evidence supplied by the studies provisionally eligible for the *primary synthesis* and relevant methodological developments discussed in *supporting literature.* At the present screening stage, five studies are provisionally eligible: R04 [17], R12 [1], R19 [8], R20 [18], and R22 [14]. R25 [5] remains unresolved because the provenance of its reported log inputs has not been established, and R26–R28 [10, 19, 20] have not yet been screened. Accordingly, the observations below are an evidence-based description of

the current candidate set, rather than a bibliometric estimate of the entire field or a final chronological trend.

3.4.1 Publication Coverage and Temporal Interpretation

The provisional primary studies were published in 2023, 2025, and 2026. R22 [14] (2023) compares traditional syntactic representations and Transformer-derived semantic features for unsupervised anomaly detection on web and application data. Three studies from 2025 illustrate distinct analytical directions: R04 [17] investigates distilled graph-based learning on host audit provenance; R19 [8] applies an attention-enhanced LSTM to application-log anomalies; and R20 [18] examines an attention-enhanced LSTM autoencoder for authentication-sequence anomalies. R12 [1] (2026) presents an integrated monitoring architecture combining log analysis, large language models, retrieval-supported interpretation, alerting, and automated workflows.

Table 10 organizes these studies chronologically and identifies the methodological contribution illustrated by each publication. The table is a *milestone map*, not a frequency analysis of all publications from 2020 to 2026. In particular, the absence of provisionally eligible studies from certain years must not be interpreted as an absence of research activity in those years. Similarly, the 2024 arXiv version associated with R04 [17] is not counted separately from its 2025 conference publication.

The available chronology is therefore better interpreted as evidence of *methodological coexistence* than as a linear replacement of traditional ML by deep learning and, subsequently, by LLMs. In 2025 alone, the candidate set includes graph-based, supervised recurrent, and reconstruction-based approaches addressing different kinds of security evidence. Their simultaneous use demonstrates that method selection remains closely tied to the characteristics of the log source and the intended security objective.

3.4.2 Changes in Log Representation: From Handcrafted Features to Contextual and Relational Evidence

A central dimension of methodological development concerns the representation of security events. Traditional log analytics often extract explicit tokens, templates, counts, or other features that can be processed by statistical and machine-learning algorithms. Such representations can be computationally economical and comparatively straightforward to inspect, but they may omit contextual distinctions between superficially similar event messages. Transformer-based embedding methods offer a different representation strategy by encoding richer semantic information from log content, although the resulting representations can be more expensive to compute and less transparent to interpret.

R22 [14] provides a particularly relevant comparison because it evaluates both representation strategies within the same experimental study. Its use of TF–IDF and Dis-

Table 10: Chronological methodological milestones in the provisionally eligible studies

Year	**Study**	**Methodological emphasis**	**Evidence boundary**
2023	R22 [14]	Syntactic features versus Transformer-derived semantic embeddings with unsupervised detectors	Results vary by dataset and algorithm; semantic features carry additional computational cost.
2025	R04 [17]	Relational audit-provenance modeling and graph-model knowledge distillation	Attack-oriented benchmark evidence does not by itself establish general deployment effectiveness.
2025	R19 [8]	Attention-enhanced recurrent classification of application-log anomalies	Aggregate results combine security-relevant and operational categories with rule-derived labels.
2025	R20 [18]	Attention-enhanced sequence reconstruction of organizational sign-in events	Behavioral anomaly labels do not independently verify malicious authentication.
2026	R12 [1]	LLM-assisted log monitoring, retrieval-supported interpretation, alerting, and workflow integration	Primarily qualitative evidence; attack-detection and security-response improvements are not quantitatively established.

Note: Years refer to the publications used for screening; R04 [17]'s 2024 preprint and 2025 conference version represent one research contribution. Study membership is provisional. This table is not intended to support publication-growth statistics or a claim that techniques developed in a strictly sequential order.

tilRoBERTa embeddings with multiple unsupervised learning algorithms shows that the choice of representation cannot be separated from the dataset or classifier. Semantic features improve some reported combinations, whereas syntactic representations perform better or similarly in others. The study also reports substantially longer feature-extraction times for semantic embeddings. These findings challenge a simplistic evolutionary account in which increasingly sophisticated representations consistently supersede earlier ones.

Another development concerns relationships among events. Host audit data can describe interactions among processes, files, and other system objects. These interactions may form a more informative unit of analysis than isolated messages. R04 [17] uses relational representations derived from host audit provenance and combines graph-based learning with knowledge distillation. In this setting, architectural development is directed not primarily toward natural-language interpretation but toward modeling attack-relevant relationships while controlling model complexity. This illustrates that advances in security log representation may involve *structural context* as well as *semantic context*.

The distinction is consequential for cybersecurity research. A semantic embedding can help characterize the meaning of a request or log message, while an audit-provenance graph can represent which entities interacted and how activity propagated. Neither type of representation is intrinsically superior for every task. Their appropriateness depends on the recorded events, the relevant attack behaviors, and the evidence needed to support an analyst's decision.

3.4.3 Development of Learning Objectives and Behavioral Modeling

The candidate studies also differ in how they formulate learning objectives. Some approaches aim to classify predefined event categories, whereas others characterize normal behavior and identify deviations. These formulations have different implications for the availability of labels, the handling of previously unseen patterns, and the interpretation of detection outputs.

In R19 [8], an attention-enhanced LSTM distinguishes three application-log categories: normal, user anomalies, and system anomalies. Its objective is explicitly discriminative: the model learns to predict categories defined by the dataset's labeling procedure. R20 [18], by contrast, employs an attention-enhanced LSTM autoencoder to reconstruct sequences of Microsoft Entra ID sign-in events, with deviations indicated by reconstruction error. These models use related neural building blocks, yet the meanings of their outputs differ. A predicted user-anomaly class and a large sequence-reconstruction error do not establish the same security claim.

The coexistence of these approaches demonstrates that recurrent modeling has not evolved toward a single dominant formulation. Instead, classification and reconstruction-

based anomaly detection address distinct data and labeling constraints. Furthermore, the labels used in R19 [8] and R20 [18] are partly derived from rules or behavioral heuristics. As a result, improvements in their respective classification or anomaly scores should not be interpreted as independently validated improvements in detecting real intrusions or account compromise. In this respect, the validity of the target labels remains as important as the development of the model architecture.

R04 [17] illustrates another modeling objective: learning to discriminate attack-related behavior in relational host activity while transferring information from a more complex graph-based model to a lighter student model. This emphasis on knowledge distillation reflects a practical concern with deployability, but the resulting efficiency claims must be interpreted within the reported datasets and configurations. The development of model architectures thus encompasses not only predictive ability but also trade-offs among context preservation, computation, and operational constraints.

3.4.4 From Automated Detection Toward LLM-Assisted Interpretation and Workflows

A further development is the use of language models to assist with the interpretation of machine-generated events. Detection-oriented systems generally output anomaly scores, categories, or alerts. LLM-assisted pipelines may additionally summarize events, explain possible causes, organize investigation context, or propose actions. These outputs may be valuable to security analysts, but their evidential status differs from that of experimentally validated attack-detection decisions.

R12 [1] provides a later example in the current candidate set. Its architecture combines ELK-based log monitoring with LLM-assisted analysis, retrieval-supported contextualization, alerts, and automated workflows. The study demonstrates the feasibility of joining components that were previously often assessed as separate tasks. However, its predominantly qualitative evaluation does not establish that the integrated system detects attacks more accurately or resolves incidents more safely than alternatives. The presence of workflow automation must therefore not be equated with verified autonomous security response.

Supporting literature helps contextualize this architectural direction without enlarging the primary evidence base. For example, R23 [9] (LogRAG, 2024) combines a one-class log anomaly detector with retrieval-augmented LLM reassessment of suspected anomalous messages. Its reported evaluation concerns operational BGL log anomalies rather than cybersecurity-specific attacks. Likewise, R24 [3] (2026) integrates topology-aware log–metric correlation and LLM-based root-cause interpretation but evaluates an induced GPU out-of-memory failure rather than a security incident. These studies demonstrate relevant technical developments in retrieval and explanation, yet their fault-oriented re-

sults cannot be treated as measured improvements in threat detection. They are discussed here only as methodological context, consistent with the study-selection rules defined in Chapter 2.

The broader implication is that the functional scope of some log-analysis systems has expanded from producing alerts toward generating contextual interpretations and supporting downstream actions. Nevertheless, this should be described as a *change in architectural scope*, not as proof of a transition to reliable autonomous cybersecurity. Such a conclusion would require controlled evaluations of response correctness, false-alert consequences, analyst workload, latency, privacy, and remediation safety under genuine security scenarios.

3.4.5 Synthesis of Temporal Developments and Implications for RQ4

Taken together, the studies provisionally included in this review reveal several developments relevant to RQ4. First, the available evidence encompasses multiple representations of log information, including syntactic features, Transformer-derived embeddings, event sequences, and host-provenance graphs. Second, models differ in their learning objectives, ranging from discriminative classification and reconstruction-based anomaly scoring to attack-oriented graph analysis. Third, later integrated pipelines illustrate the use of LLMs and retrieval mechanisms for contextual explanation, alert interpretation, and workflow support.

These observations do *not* establish a universal chronological progression in which one paradigm replaces another. The methods overlap in publication time and address heterogeneous data sources and security objectives. Nor does the candidate set support a reliable estimate of publication growth or the relative prevalence of each AI paradigm across the entire 2020–2026 period. Such claims would require the final screened corpus and an explicitly defined counting procedure.

More importantly, the identified developments have not eliminated recurring methodological limitations. Semantic representations can incur substantial processing costs; behavior-based labels may not correspond to independently verified attacks; security benchmarks may not represent deployed systems; and LLM-generated analyses may lack controlled measures of correctness and response effectiveness. Thus, the evolution of AI-based security log analysis is best characterized, within the present evidence, as a diversification of *representations, learning objectives, and system functions* accompanied by persistent challenges in evaluation validity and operational trustworthiness.

The discussion of Research Question RQ4 will be updated when screening and QA1–QA6 quality appraisal are completed. The next section examines which research tendencies recur across the eligible evidence and separates these tendencies from capabilities documented only in the supporting literature.

3.5 Research Trends in AI-Based Security Log Analysis

The methodological evolution described in Section 3.4 also reveals several cross-cutting research tendencies. While the previous section considered the development and coexistence of AI approaches over time, the present section examines recurring analytical priorities that connect studies using different models and log sources. The distinction is important: a chronological account identifies when a particular approach appears in the reviewed literature, whereas a trend-oriented synthesis considers the research problem that motivates the approach and the extent to which the reported evidence supports its intended contribution.

The analysis addresses the trend-oriented component of Research Question RQ4 and provides an evidence-based bridge to the research gaps discussed in Section 3.6. At this stage, the provisional primary evidence consists of R04 [17], R12 [1], R19 [8], R20 [18], and R22 [14]. Study R25 [5] remains unresolved because the provenance of its experimental input records has not been sufficiently established, and R26–R28 [10, 19, 20] await eligibility screening. Consequently, the following themes are *qualitative tendencies observable in the current candidate set*, not statistically estimated frequencies, field-wide rankings, or claims that all techniques have been tested under comparable conditions. Methodologically relevant but ineligible studies are identified explicitly as supporting literature; their operational-anomaly results are not interpreted as cybersecurity attack-detection evidence.

3.5.1 Context-Aware and Structure-Preserving Representations

One recurring priority is the preservation of event context during the transformation of heterogeneous logs into machine-processable representations. Security evidence can be weakened when preprocessing treats complex events as independent tokens or discards relations among entities. Conversely, retaining additional semantics or structure introduces computational costs and may not improve detection in every setting. Thus, the emerging concern is not simply to construct richer embeddings but to identify which contextual information contributes to a specific security decision.

R22 [14] provides a direct comparison of syntactic and Transformer-derived semantic features for unsupervised anomaly detection on web and application data. The study reports that the relative usefulness of semantic embeddings depends on both the dataset and the downstream detection algorithm: some experimental configurations favor semantic features, whereas others favor simpler syntactic representations. It also reports greater extraction time for semantic features. These findings qualify any claim that contextual embedding methods universally outperform conventional representations. They suggest instead that representation design and model choice must be evaluated jointly against the target log source and security task.

R04 [17] illustrates a different meaning of context preservation. It uses host-audit provenance to model relationships among operating-system entities and applies graph-based knowledge distillation for attack-oriented detection. Here, the analytical value arises from interactions among processes, files, and related system objects rather than primarily from the linguistic meaning of individual event messages. The contrast between R04 and R22 [14] indicates that *semantic context* and *relational context* are complementary but non-interchangeable forms of security evidence. A broader research opportunity is to determine when such context can be preserved without imposing prohibitive data-collection and inference costs.

3.5.2 Behavioral Modeling: Anomalies Versus Threats

Another prominent concern is the modeling of behavior rather than reliance solely on isolated indicators. Behavioral approaches seek to describe ordinary user, application, or system activity and identify departures from the learned or predefined pattern. Their practical appeal is that unusual behavior may be observable even when a familiar attack signature is unavailable. However, the interpretation of an anomaly depends on how normality is defined and on whether the evaluation distinguishes benign deviations from confirmed malicious activity.

R20 [18] investigates attention-enhanced reconstruction of Microsoft Entra ID sign-in sequences, using reconstruction error to detect unusual authentication behavior. Its reported anomaly-class F1-score of 0.81, compared with 0.80 for its LSTM-autoencoder baseline, reflects performance against behavioral anomaly labels derived from heuristics rather than independently verified malicious logins. The result therefore supports a limited claim about anomaly recognition in the evaluated authentication data, not a measured success rate for account-compromise detection. R19 [8] addresses a related but different task by classifying application logs into normal, user-anomaly, and system-anomaly categories with an attention-enhanced LSTM. Although user anomalies include security-relevant behavior, the study reports aggregate results across security-related and operational categories, with labels derived from defined rules.

These two studies illustrate the need to separate *behavioral anomaly detection*, *security alert generation*, and *validated attack detection* in the synthesis. They also show that closely related neural architectures can serve distinct learning objectives: a discriminative classifier predicts predefined categories, whereas a reconstruction model assigns anomaly scores relative to a learned behavioral baseline. Future comparisons therefore require explicit documentation of label provenance, evaluation unit, threshold selection, and the presence or absence of independently validated incident outcomes.

3.5.3 Efficiency, Computational Cost, and Deployment Constraints

A third tendency is increasing attention to the cost of processing security-relevant logs in realistic environments. Large event volumes, continuous collection, and distributed data sources create constraints that cannot be captured by predictive performance metrics alone. For operational use, relevant considerations include the latency of representation extraction, model size, resource utilization, and the ability to process incoming events without excessive delay.

R04 [17] addresses computational efficiency through knowledge distillation from a graph-based teacher to a lighter student model. The proposed design aims to retain attack-relevant relational knowledge while reducing the burden of graph-based threat analysis. However, published efficiency results remain tied to the specified benchmarks and configurations and should not be generalized to untested deployment environments. R22 [14] supplies a complementary perspective by measuring the additional processing time associated with Transformer-derived semantic representations. Its findings demonstrate an explicit trade-off between richer features and extraction cost, with no consistently superior representation across every evaluated dataset and model.

Taken together, these approaches suggest a shift in research emphasis toward *task-appropriate efficiency* rather than architectural complexity alone. The supporting evidence does not establish a common cost-performance frontier because the studies evaluate different tasks, hardware contexts, and computational stages. It does, however, motivate more consistent reporting of preprocessing time, throughput, memory requirements, inference latency, and operational constraints in security-log evaluations.

3.5.4 LLM-Assisted Interpretation and Retrieval-Supported Analysis

A further tendency is to extend log-analysis systems beyond numerical classifications or anomaly scores toward explanations, summaries, and contextual assistance for analysts. Large Language Models (LLMs) can generate natural-language interpretations of retrieved events and facilitate interaction with complex monitoring outputs. Retrieval-Augmented Generation (RAG) can provide additional event or knowledge context during such interpretation. Nonetheless, the generation of a plausible explanation does not itself establish diagnostic correctness, attack attribution, or reduced incident-response time.

R12 [1] examines this direction through an integrated ELK-based monitoring pipeline with LLM-assisted log interpretation, retrieval-supported components, alerting, and automated workflows. Its OpenSSH examples provide security-relevant operational context. The experiments primarily demonstrate integration and qualitative behavior, however, rather than a controlled improvement in attack-classification accuracy, analyst performance, or response quality. R12 therefore provides evidence of architectural feasibility, but not of the reliability of every generated explanation or automated action.

Supporting studies illustrate the wider technical interest in contextual retrieval without enlarging the primary security evidence base. R23 [9] (LogRAG) combines a one-class detector with retrieval-augmented LLM reassessment and reports reduced false positives on operational BGL log anomalies. R24 [3] couples topology-aware log–metric correlation with LLM-assisted root-cause interpretation, but validates the approach using an induced GPU out-of-memory fault. Both contributions are relevant to the design of retrieval and diagnostic pipelines; neither supplies direct evidence of improved cybersecurity threat detection under the inclusion criteria of this review.

This distinction suggests two complementary research needs: verifying that retrieved material is relevant and faithful to the source events, and assessing whether generated conclusions are accurate, appropriately uncertain, and useful to analysts. Evidence of concise or readable output should not be substituted for measurements of factual grounding, false-positive consequences, investigation time, or analyst decision quality.

3.5.5 Integration of Detection, Alerting, and Security Workflows

Some studies also seek to embed AI analysis within broader monitoring and action pipelines rather than treating a detector as a self-contained component. Such integration may connect log ingestion, anomaly identification, contextual interpretation, alert routing, and operational actions. In principle, linking these functions could reduce the effort needed to move from an observed event to an investigation decision. In practice, the safety of this integration depends on the accuracy of intermediate outputs and on the controls governing downstream actions.

R12 [1] provides the clearest provisional example of this systems-level objective. Its architecture connects log monitoring and LLM-assisted interpretation with alerts and workflow-oriented automation. Nevertheless, the presence of automated jobs or self-healing components in a prototype is not equivalent to experimentally validated *autonomous cybersecurity incident response*. The study does not establish, through controlled attack scenarios, that generated decisions lead to correct containment or remediation without unacceptable unintended effects.

Accordingly, the trend is best characterized as movement toward *workflow-connected security log analysis*, not demonstrated end-to-end autonomy. Claims of autonomous response would require evaluation of action correctness, human-approval boundaries, rollback and recovery mechanisms, false-alert consequences, and measurable improvements in containment or investigation outcomes. These requirements form part of the research gap analysis developed in Section 3.6.

3.5.6 Evidence Quality as a Cross-Cutting Research Concern

The provisional studies also reveal a recurring methodological tension between technically promising demonstrations and the strength of their cybersecurity-specific evaluation. This should not be described as a measured rise in evaluation quality over the publication period; rather, it is an analytical concern that becomes visible when heterogeneous findings are placed side by side.

R04 [17] assesses an explicitly attack-oriented host-audit task, whereas R20 [18] measures heuristic authentication anomalies and R19 [8] reports overall classification across a mixture of user-related and operational log anomalies. R22 [14] uses attack-containing web and HTTP request benchmarks but also reports dataset-dependent results and tuning choices that limit purely unsupervised interpretation. R12 [1] offers qualitative evidence for an integrated LLM-assisted monitoring workflow without a standardized, labeled benchmark of attack-detection or remediation outcomes. Each study contributes evidence for a particular task, but the studies do not establish a single comparable scale of practical security effectiveness.

For this reason, the present review treats *dataset provenance, attack-ground-truth validity, label construction, evaluation granularity, and reproducibility* as essential conditions for interpreting emerging approaches. Research on operational log anomalies may offer useful algorithms and pipeline designs; nevertheless, transferring its performance claims into a cybersecurity setting requires separate, security-specific evaluation. This boundary remains especially important for new language-model-assisted pipelines, whose output fluency can obscure unsupported inferences.

Table 11 consolidates the identified themes, the studies that illustrate them, and the limits of their available evidence. The table provides a thematic evidence map rather than a quantitative frequency ranking: a study may contribute to multiple themes, and supporting studies are not counted as primary security validation.

3.5.7 Synthesis of Research Trends and Implications for RQ4

The research themes identified in the current candidate set reflect a diversification of the questions addressed by AI-based security log analysis. Some approaches attempt to retain richer semantic or relational evidence, others model sequences of user or application behavior, and more recent systems experiments connect detection outputs with language-model-based interpretation and workflow components. Efficiency constraints and the reliability of security labels remain relevant across these approaches. These observations are consistent with an expansion of methodological possibilities but do not establish that any single paradigm has become dominant or uniformly more effective.

In relation to RQ4, the principal trend is therefore a growing *breadth of analytical functions* rather than a demonstrated linear progression toward autonomous defense.

Table 11: Cross-cutting research themes and their evidentiary boundaries

Research theme	**Primary examples**	**Supported observation**	**Main evidence limitation**
Context-enriched representations	R04 [17], R22 [14]	Relational audit graphs and semantic or syntactic features support different analytical tasks.	No representation is consistently superior across heterogeneous datasets.
Behavioral modeling	R19 [8], R20 [18]	Classification and reconstruction models detect different kinds of anomalous activity.	Rule-derived anomalies do not necessarily correspond to verified attacks.
Efficient model deployment	R04 [17], R22 [14]	Distillation and representation-cost analysis make resource constraints explicit.	Efficiency results are not reported under a common operational benchmark.
LLM and retrieval support	R12 [1]	Contextual interpretation and retrieval can be integrated into monitoring pipelines.	Limited controlled evaluation of security correctness and analyst benefits.
Workflow-connected analysis	R12 [1]	Alerts and automation can be coupled to log analysis and LLM outputs.	End-to-end security response safety and effectiveness remain unverified.

Note: Themes were derived qualitatively from provisionally eligible primary studies and are not mutually exclusive or frequency-ranked. R23 [9] and R24 [3] are discussed in the text only as supporting methodological literature because their published evaluations address operational anomalies or faults. R25 [5] and R26–R28 [10,19,20] are excluded from the present thematic evidence map pending eligibility review. All conclusions remain subject to QA1–QA6 appraisal.

Models that identify anomalous events, systems that explain those events, and systems that execute follow-up actions must be evaluated against distinct outcome measures. Their roles may be complementary, but improvements in one role cannot automatically be transferred to another. The present evidence also indicates that log-source characteristics and experimental ground truth remain central to the practical interpretation of reported gains.

The conclusions in this section are provisional because the eligibility review and methodological quality assessment are still in progress. Once the primary corpus is finalized, the thematic map should be re-examined to determine whether additional qualifying studies strengthen, qualify, or contradict the tendencies reported here. Section 3.6 extends this evidence-based synthesis by identifying the unresolved research gaps and specifying the kinds of security-focused evaluation needed to address them.

3.6 Research Gaps and Future Directions

The classification, comparative analysis, and thematic synthesis presented in the preceding sections identify several limitations that remain relevant to the development of Artificial Intelligence (AI)-based security log analysis. Although the reviewed approaches demonstrate different ways of representing security events, identifying unusual behavior, and assisting analysts, the strength of their conclusions depends on the security tasks, log sources, labeling procedures, and experimental settings used for evaluation. An important purpose of this review is therefore to distinguish an unresolved *security research problem* from a limitation that applies only to an individual implementation.

The gaps discussed below address Research Question RQ5 by connecting observations from the provisionally eligible primary studies to specific requirements for future research. At the present stage, the candidate primary set comprises R04 [17], R12 [1], R19 [8], R20 [18], and R22 [14]. R25 [5] remains subject to clarification of its input-data provenance, R26–R28 [10,19,20] have not yet completed eligibility screening, and the final quality assessment (QA1–QA6) is pending. Consequently, the discussion identifies *gaps evidenced by the currently reviewed studies*; it does not establish their prevalence across all publications in the field. Supporting studies are used only where they clarify transferable methodological approaches or evaluation concerns, not as independent evidence of cybersecurity attack-detection effectiveness.

3.6.1 Security Ground Truth and Label Validity

A central challenge is the distinction between events labeled as anomalous and events independently established to be malicious. Many log-analysis methods learn a representation of normal behavior and identify observations that differ from this representation. This can be useful for alert triage, but it does not necessarily reveal whether the un-

derlying event represents an attack, an authorized administrative action, a configuration change, or an ordinary operational failure.

This distinction is apparent in R20 [18], where authentication anomalies are evaluated using rule-derived behavioral labels rather than independently confirmed account-compromise incidents. The reported anomaly-detection results therefore characterize agreement with the labeling rules, not a verified attack-detection rate. R19 [8] similarly evaluates application logs containing both user-related and system-related anomalies, with labels derived from predefined rules. The study's aggregate results do not isolate the model's ability to detect separately validated malicious events. By contrast, R04 [17] evaluates an explicitly attack-oriented host-audit task, illustrating the added interpretive value of security-specific scenarios, although the evaluation remains bounded by the selected datasets and attack cases.

Future studies should distinguish among *operational anomalies*, *security-relevant but unconfirmed anomalies*, and *verified malicious events* in both dataset construction and performance reporting. Stronger validation could combine controlled attack execution, documented incident records, and independent expert assessment. Where complete ground truth is unavailable, uncertainty in labels should be disclosed instead of being treated as an unqualified binary distinction between benign and malicious behavior.

3.6.2 Insufficient Evidence of Generalization Across Environments and Time

The effectiveness of an AI-based log-analysis approach can depend heavily on the structure of the log source, the organization generating the records, the attack scenarios represented in the evaluation, and changes in application behavior over time. A model that performs well within a particular benchmark may behave differently when deployed on new software versions, organizations, user populations, or previously unseen attack techniques.

R22 [14] makes this dependence especially visible by comparing syntactic and semantic features across different HTTP and application datasets. Its findings show that the relative performance of these representations varies with both the dataset and the downstream algorithm. The study also includes configuration choices informed by available labels, which must be considered when interpreting the degree to which the procedure is independent of labeled evaluation data. R20 [18], meanwhile, evaluates authentication behavior over a limited organizational observation period, leaving the transferability of its findings to other identity environments unresolved. R04 [17] provides attack-oriented evaluation on provenance datasets, but the results do not alone establish performance under all production logging configurations.

Future research should employ temporal holdout evaluation, cross-environment testing, and explicit assessment of unseen or evolving attack behavior. Where feasible, studies

should test models on data collected after the training period and from an organization or deployment not represented during development. Hyperparameter selection, threshold calibration, and representation learning should be separated from final test data to minimize optimistic estimates. Researchers should also report how detection performance changes as log formats, authentication practices, and application workloads evolve.

3.6.3 Fragmented Evaluation Practices and Limited Cross-Study Comparability

A further methodological gap concerns the absence of a consistent basis for comparing reported outcomes. The provisionally included studies evaluate different prediction targets using different units of analysis and performance metrics. Consequently, apparently similar numerical scores may represent substantially different capabilities.

For example, R19 [8] reports an overall F1-score of 98.55% for a mixed application-log anomaly classification task, while R20 [18] reports an anomaly-class F1-score of 0.81 on sign-in data. R22 [14] presents weighted F1-scores for specific web and application benchmark configurations, and R12 [1] offers predominantly qualitative evidence for an integrated monitoring and LLM-assisted workflow. None of these outputs should be directly interpreted as measuring the same rate of detecting confirmed attacks. The differences are not merely numerical: they concern the definition of the positive class, the provenance of labels, the granularity of prediction, and the purpose of the system.

Future empirical studies should report the unit of evaluation (event, session, entity, alert, or incident), class distributions, labeling procedures, data-partitioning strategies, and threshold-selection methods. Per-class precision and recall, confusion matrices, and appropriately interpreted F1-scores should be reported when relevant; precision–recall analysis and false-alert rates are particularly important for infrequent security events. For analyst-facing systems, additional outcomes may include the number of alerts requiring review, the time required to investigate an event, and the cost of false positives. Comparative claims should be restricted to genuinely comparable tasks and experimental conditions rather than aggregated across incompatible datasets.

3.6.4 Incomplete Correlation of Heterogeneous Security Evidence

Security incidents may unfold across multiple hosts, applications, identities, and network services, while the available log records often provide only partial observations of those activities. Approaches that analyze a single source can detect useful local indicators but may lack sufficient context to connect events into an evidence-supported account of an incident. Conversely, combining many sources introduces challenges in event identity, timestamp alignment, missing records, inconsistent schemas, and causal interpretation.

The current provisional evidence demonstrates distinct, source-specific strengths.

R04 [17] preserves host-level entity relationships in provenance graphs, R20 [18] analyzes authentication sequences, and R22 [14] examines HTTP and application request representations. However, these studies do not collectively establish a validated, end-to-end method for linking identity, host, and web events into a single incident narrative under common evaluation conditions. Operational research on topology-aware log–metric correlation, such as R24 [3], illustrates a possible technical component for contextualizing distributed events, but its GPU fault-diagnosis evaluation is not evidence of cybersecurity incident correlation.

A useful research direction is the development of security-specific correlation methods that integrate heterogeneous event sources while retaining explicit provenance links. Such methods should be assessed against multi-stage attack scenarios and distinguish temporal co-occurrence from defensible causal relationships. Evaluations should include missing or delayed records, ambiguous entity identifiers, inconsistent timestamps, and competing benign explanations. They should also assess whether source integration improves incident reconstruction or analyst decisions beyond what is achieved by individual sources.

3.6.5 Limited Validation of Explanations and Retrieval-Augmented Reasoning

The introduction of attention mechanisms, graph-based representations, and Large Language Models (LLMs) creates new opportunities for interpreting suspicious events. Yet an output that appears understandable is not necessarily faithful to the evidence or useful for making a security decision. The methodological challenge is to evaluate both the predictive result and the validity of the accompanying explanation.

In R19 [8] and R20 [18], attention-related components form part of the neural architecture, but this does not independently establish that learned attention weights correspond to the true causes of a security event. R04 [17] provides relational evidence that may assist interpretation of host activity, yet relationships in a provenance graph should not automatically be equated with an independently verified causal explanation of every detected event. R12 [1] demonstrates the integration of LLM-assisted interpretation and retrieval-supported processing into log monitoring, but its largely qualitative evaluation does not establish the factual correctness or investigative value of generated security explanations.

Supporting study R23 [9] provides a relevant technical example of retrieving normal log templates to reassess anomalies using an LLM; however, its reported experiments concern operational BGL anomalies rather than attacks. Therefore, its reduction of operational false positives cannot be transferred directly to security-alert precision. These examples motivate the separation of three distinct properties: *detection correctness*, *evi-*

dence grounding, and *explanation usefulness*.

Future work on LLM- and retrieval-assisted security analysis should evaluate whether generated statements are traceable to identifiable log records, whether retrieved evidence is relevant, and whether the system appropriately indicates uncertainty when the evidence is insufficient. Security-focused test scenarios should include contradictory records, incomplete context, unfamiliar event formats, and misleading or attacker-controlled text within log fields. In addition to automated language-quality measures, evaluations should assess unsupported claims, missed evidence, and analyst decision quality. Human review or independent verification remains important before generated interpretations are used as the basis for consequential security actions.

3.6.6 Underreported Efficiency, Deployment, and Privacy Requirements

Detection quality is only one dimension of operational feasibility. Security monitoring systems may need to process high-volume logs under latency constraints while handling sensitive information about accounts, devices, internal infrastructure, and organizational behavior. The use of large encoders, provenance graphs, external model services, and retrieval pipelines can create different resource and governance requirements, none of which is adequately captured by accuracy or F1-score alone.

R04 [17] considers efficiency through knowledge distillation in graph-based threat detection, while R22 [14] explicitly reports the additional representation-extraction cost associated with semantic embeddings relative to simpler syntactic features. These studies illustrate practical trade-offs but do not offer a shared deployment benchmark. R12 [1] demonstrates integrated log monitoring and LLM-assisted processing, yet its evaluation does not comprehensively quantify processing latency, sustained throughput, cost, or secure handling of sensitive log content in production. Authentication logs such as those analyzed in R20 [18] further illustrate why access to identity-related records requires careful governance.

Future evaluations should report the computational and operational conditions under which a security model is tested, including processing throughput, preprocessing and inference latency, memory consumption, model or service cost, and behavior under peak log volumes. Privacy-preserving processing, field minimization, appropriate pseudonymization, access control, retention limits, and auditability should be treated as deployment requirements rather than assumed consequences of using AI. When an external LLM service is involved, studies should clarify what security-relevant data leave the organizational boundary and how that transfer is controlled. The appropriate safeguards will depend on the deployment context and should be evaluated rather than asserted.

3.6.7 Limited Evidence for Safe and Effective Automated Incident Response

Moving from the detection of suspicious activity to automated incident response introduces a distinct set of research problems. A detector may correctly raise an alert without providing sufficient information to justify account suspension, host isolation, rule changes, or automated remediation. The consequences of an incorrect action can be more serious than those of an incorrect classification alone, particularly in critical or highly available systems.

R12 [1] connects security log monitoring with alerts and workflow-oriented automation, demonstrating an architectural direction toward more integrated operations. However, the published evidence does not establish, under controlled cybersecurity incidents, that automatically generated actions achieve correct containment or remediation without unacceptable operational consequences. Likewise, work on self-healing operational systems cannot be treated as validated security incident response merely because the underlying automation is effective for system faults.

Research on autonomous or semi-autonomous response should therefore evaluate the entire decision chain: the initial evidence, the inferred security risk, the selected action, the authorization boundary, and the resulting security and operational outcomes. Relevant measures include containment correctness, time to investigation or containment, reversibility, service disruption, and the cost of inappropriate actions. High-impact actions should be assessed under clearly defined human oversight or policy constraints, with mechanisms for audit trails, rollback, and independent verification of outcomes. These requirements are prerequisites for defensible claims of safe autonomous security response.

3.6.8 Evidence-Linked Future Research Agenda

The preceding gaps are interconnected. Improving threat-ground-truth quality strengthens the interpretation of predictive performance; rigorous evaluation across time and environments supports conclusions about generalization; and evidence-preserving correlation provides a stronger basis for reliable explanations and response decisions. Consequently, future research should not treat improved benchmark scores as a sufficient substitute for end-to-end security validation.

Table 12 consolidates the research gaps supported by the provisional evidence and the corresponding directions for investigation. The table is a qualitative evidence map, not a quantitative priority ranking. Study IDs identify papers that illustrate each concern; they do not imply that the limitation applies equally to every study in the literature.

The table suggests that improving the quality of security evidence is a prerequisite for many subsequent advances. For example, a system cannot reliably demonstrate an improvement in attack detection if its positive class is defined only by unverified behavioral heuristics. Similarly, a generated incident explanation cannot be considered dependable

Table 12: Evidence-linked research gaps and recommended future directions

Research gap	**Illustrative studies**	**Recommended research direction**
Ambiguous security ground truth	R19 [8], R20 [18]; contrast R04 [17]	Separate behavioral anomalies from verified attacks; document labeling and uncertainty.
Limited temporal and cross-domain validation	R04 [17], R20 [18], R22 [14]	Use chronological and cross-environment tests with unseen attacks and isolated tuning data.
Incompatible tasks and metrics	R12 [1], R19 [8], R20 [18], R22 [14]	Standardize outcome definitions, evaluation units, per-class metrics, and false-alert reporting.
Fragmented multi-source evidence	R04 [17], R20 [18], R22 [14]; R24 [3] (supporting)	Evaluate provenance-preserving cross-source correlation on multi-stage security incidents.
Unverified explanation and retrieval quality	R12 [1], R19 [8], R20 [18]; R23 [9] (supporting)	Measure evidentiary grounding, unsupported claims, retrieval relevance, and analyst usefulness.
Insufficient operational and privacy evaluation	R04 [17], R12 [1], R20 [18], R22 [14]	Report latency, throughput, resource costs, data governance, and secure processing constraints.
Unvalidated response automation	R12 [1]	Test authorized, reversible incident actions with measured security and operational outcomes.

Note: The mapping reflects the current provisional primary-study set. R23 [9] and R24 [3] are cited only as supporting methodological literature because their evaluations address operational anomalies or faults. R25 [5] awaits clarification of its source-data provenance, while R26–R28 [10, 19, 20] and QA1–QA6 remain unfinished. Directions are proposed by this review; they are not claims of demonstrated effectiveness.

unless its claims can be traced to source evidence, and an automated response cannot be considered safe without evaluation of the action itself. The resulting research agenda is therefore best understood as a sequence of evidence-building requirements rather than a competition to introduce increasingly complex AI architectures.

A practical future study could address several gaps jointly by collecting time-ordered authentication, application, and host-audit events associated with documented attack and benign-activity scenarios. An evaluation could compare lightweight conventional models, context-aware neural approaches, and retrieval-supported language-model components on the same well-defined security tasks. It should assess attack-specific detection, false-alert burden, robustness to incomplete evidence, computational requirements, and the factual accuracy of any generated explanations. If response actions are included, they should be evaluated independently under explicit authorization and rollback constraints. This proposal is a research direction derived from the present synthesis, not a system implemented or experimentally validated by this review.

3.6.9 Synthesis of Research Gaps and Implications for RQ5

In answer to RQ5, the current evidence points to a set of connected methodological and operational gaps: uncertain security-specific labels, limited generalization testing, inconsistent performance reporting, incomplete integration of heterogeneous evidence, insufficient validation of explanations, underreported deployment constraints, and limited security-focused assessment of autonomous response. These gaps should not be interpreted as proof that the associated capabilities are absent from all published systems. Rather, they identify aspects that are incompletely demonstrated in the provisionally eligible evidence and consequently warrant more rigorous investigation.

The main implication is that progress in AI-based security log analysis should be assessed through the validity of security conclusions, not solely through the sophistication of the underlying model. Well-grounded evaluation requires a clear distinction between anomalies and attacks, faithful use of original log evidence, realistic deployment conditions, and security-specific measurements of both model outputs and downstream decisions. This standard applies to traditional machine learning, deep learning, graph-based methods, LLMs, and retrieval-augmented architectures alike.

Together, Sections 3.2–3.6 provide an integrated response to the research questions through a multidimensional taxonomy, comparative synthesis, analysis of temporal developments, identification of research themes, and a corresponding evidence-linked agenda. The findings remain provisional until the outstanding eligibility decisions and QA1–QA6 assessments are completed. At that point, the tables and thematic conclusions should be reconciled with the final included-study corpus and the complete bibliography before the review is submitted.

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